Introduction to Exercises

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Starting Training Applications from DOS

Make the working directory the same as the one where the process control software has been installed. To do so, use the DOS command for changing directory (CD directory name). Type "MENU" and a menu, showing available training applications will appear. Then, use the cursor keys on your keyboard to move the cursor to the training application you want to select. Press Enter and the exercise which has been selected, will come up.

Starting Training Applications from MS-Windows

Open the folder called CONTROL. Select and open the training application you want to run.

Gain of Simulated Disturbance

All training applications are configured with correct tuning constants and are ready for operation. The process simulation of each training application is configured with process disturbance process noise. The Gain for disturbance is originally set to 0 and may be changed by the student in order to add more realism into the process simulations. A value of 1 is recommended for "K-DIST" (Gain of process disturbance).

The magnitude of disturbances can be changed to challenge the operational skills of the operating student. The disturbances will create realistic situations, where operator interventions on outputs and limits are necessary to avoid loss of control. In addition, the value of the process disturbance "DISTURB" can be changed by the student directly. This permits the simulation of step functions of the process disturbance.

Example

In a level control system, a flow controller controls the inlet flow into a container. The outlet flow is unpredictable and represents the process disturbance. If the outlet flow reaches 0 and a controller output low limit (OPLO) of 5% exists, a wind-up situation exists. The container will overflow because OP is limited to 5% and no outlet flow exists. Operator intervention is required.

For most exercises, the disturbances have to be left at 0. Call up Auxiliary Display F8 for any modification to the disturbance. If only one disturbance exists in the process simulation, the disturbance gain is called K-DIST. However, if more than one disturbance is simulated, then for each disturbance exists a separate gain (K-DIST-F, K-DIST-T, K-DIST-P etc for disturbance gain of FLOW, TEMPERATURE, PRESSURE etc). Reducing the gain of the disturbance makes operation easier and less of a challenge. In order to see the principles of control clearly, no process disturbance is desired. Therefore, the exercises described below require the student to leave gain for random disturbances at 0 (default at start up).

NOTE
Stabilize the process simulation and controller before each exercise if necessary.
First, make sure disturbance gain is at 0 (Display F8). To stabilize the process, change the controller MODE to MANUAL and set the OP to 50% (in cascade control, MODE and OP of Flow Controller only). Make sure the process is steady before you start with a particular exercise.

As it is of great importance to understand the control concepts employed in different applications, the exercises will make use of the concepts explained in the previous chapters. A selection of different applications will be used to demonstrate the use of those concepts.

Trend Displays

The trend displays shown in the document are black and white only. This makes it very hard to interpret the meaning of multiple trend pens. The Trend Displays shown in the documentation have to be used together with the actual displays of the simulation.

The explanations given in the documentation are more useful when the student observes the colored trend display and the computer screen at the same time.

Exercise 1

Flow Control Loop - Basic Example

1.1 Objective

This exercise will familiarize the student with the basic concept of closed loop control. It provides an opportunity to get a first feel for closed loop control. A flow control loop will serve as a practical example for this exercise. A flow control loop is generally not critical to operate and illustrates the basic principles effectively.

1.2 Operation

Since this is a relatively simple exercise, it can be used for familiarization with the principle operation of the simulation software. Call up the training application "**Single Flow Loop**" as explained above.

Now, press the F2 button to call up the Flow Control Block Diagram Display. The display as shown in Figure 1.1 will appear. This display gives a general idea of the process and displays all major variables.

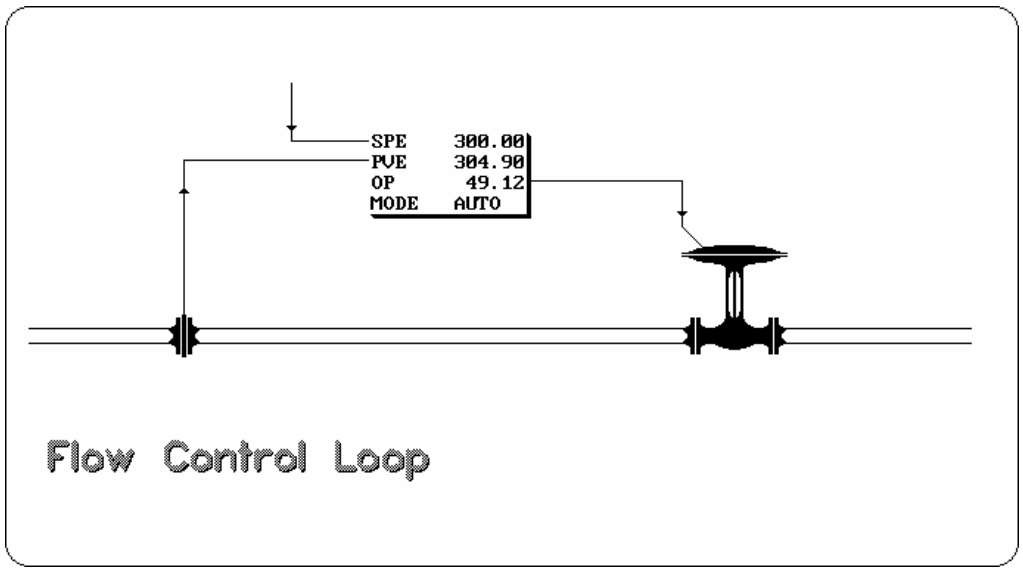


Figure 1.1
Flow Control Display

First, we will observe the general behavior of the process. This is best done on a Trend Display. Call up the Trend Display of this exercise by pressing F4. At this stage the flow control is up and running correctly in automatic control. In order to observe the process reaction, as a result of changes in the position of the control valve, change the control mode from AUTO to MANUAL. Then, change the OP-Value of the controller to 20% as follows: Type the parameter name 'OP' followed by Enter. Then enter the value 15 representing 15% of output.

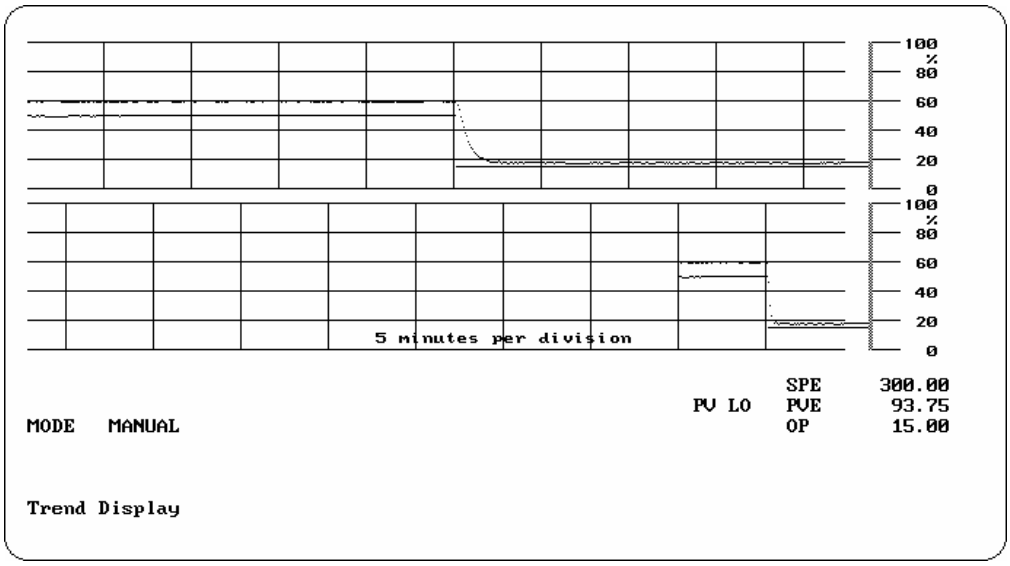


Figure 1.2
Flow Process Reaction in Manual Mode

1.3 Observation

We observe the Process Variable PVE (Process Variable in Engineering units) as it changes in value due to the change in OP. The PV will change as shown in Figure 1.2. The set point is not used in MANUAL mode and is ignored. The set point SPE (Set Point in Engineering units) just happens to be 300 at this time.

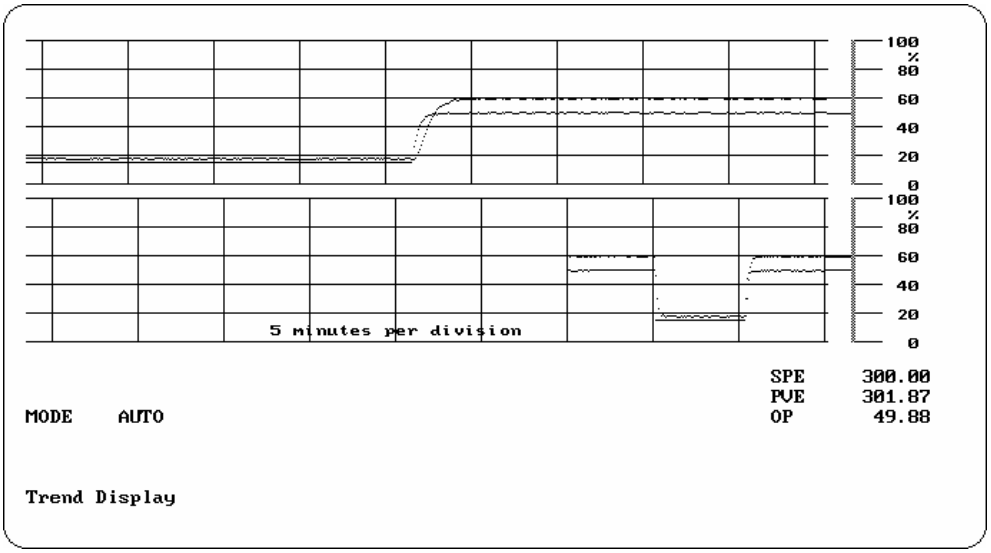


Figure 1.3
Flow Control - Automatic Return off PVE to SPE

1.4 Operation

Change the controller from MANUAL to AUTOMATIC mode.

1.5 Observation

Since the Set Point SPE is still at 300, the controller will automatically control the value of PVE to return it to the value of SPE, which is 300 (See Figure1.3).

1.6 Operation

A Detail Display can be called up with the F3 button (See Figure1.4). The student should now experiment on his/her own. You are encouraged to make changes to Range, Limits, Modes and Values in order to experience their effect. Since a flow control loop has no intrinsic stability problems, most effects can be observed clearly.

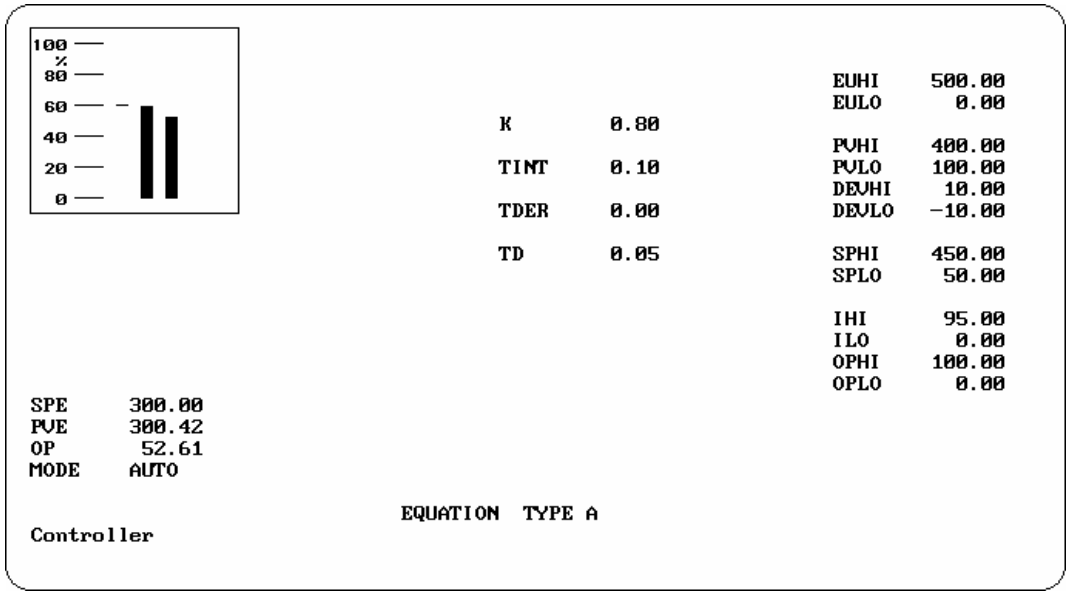


Figure 1.4
Flow Controller Detail Display

The student may call up the Auxiliary Display F8 to look at some values that are important to the simulation. The Auxiliary Display shows two columns of variables; the left column shows important background variables of the controller and the right column shows simulation variables.

Exercise 2

Proportional (P) Control- Flow Chart

2.1 Objective

This exercise will introduce the main control action of controllers - Proportional Control. Special emphasis is placed on the fact that there is a remaining offset condition, if proportional control is used solely. Figure2.1 shows an example for closed loop control.

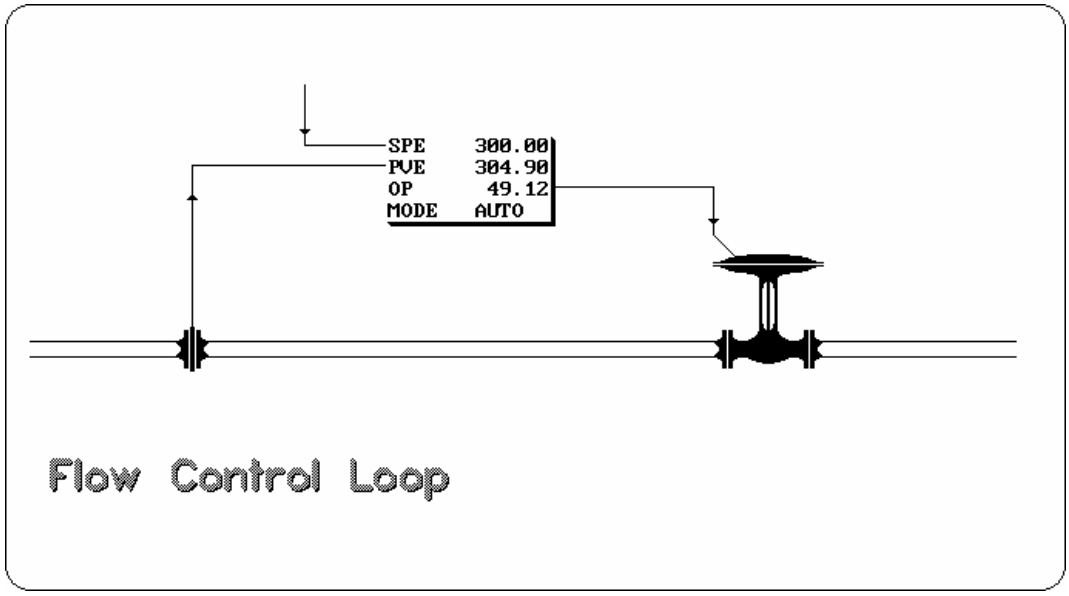


Figure 2.1
Flow Control Loop

2.2 Operation

Call up the training application "**Single Flow Loop**". After this exercise has been called up, press F3 to get the Detail Display of the flow controller.

To prepare the controller for P-Control only, change TINT (Integral Time Constant) and TDER (Derivative Time Constant). Set TINT to almost infinity. Set TDER to zero for no derivative action.

Practical values are 999 for TINT and 0 for TDER, as shown in Figure2.2.

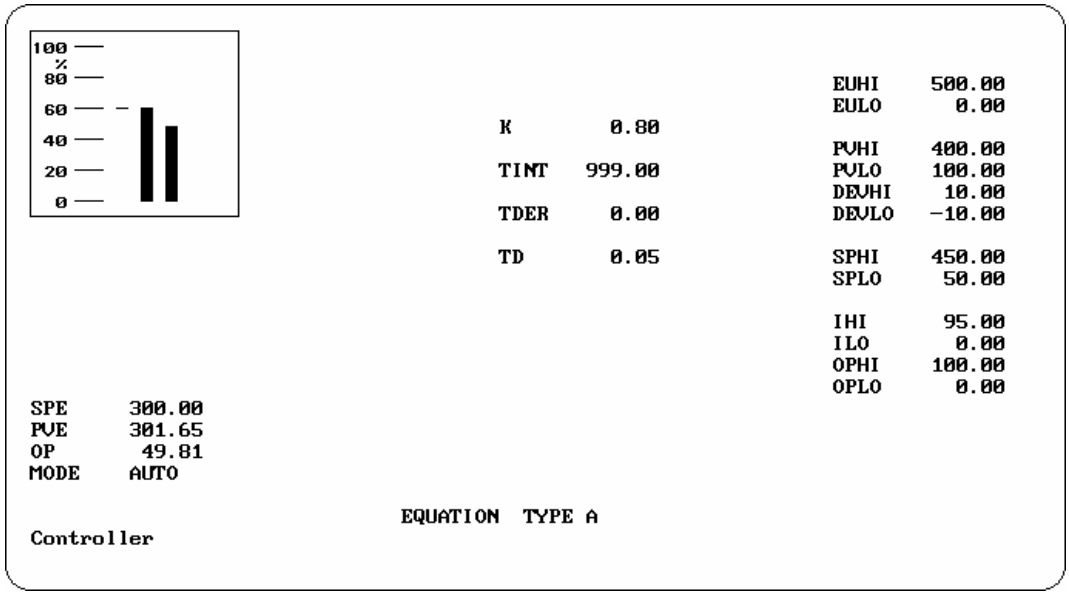


Figure 2.2
Flow Controller Detail Display for P-Control

To study P-Control, change SPE from 300 (60% of range) to 125 (25% of range).

2.3 Observation

It is observed, that the PVE value settles down before reaching the value of SPE. The remaining difference between SPE and PVE is called Proportional **Offset**. It can also be observed that the control action moves almost instantaneously (proportional control) in accordance with changes to PVE or SPE (Equation Type A). The trend display has to be observed very carefully (see Figure2.3). The step change of SPE has caused the value of OP to make a step change accordingly. Immediately after this, one can observe the output to exactly follow the value of PVE, but in reverse direction to the PVE.

Therefore, the trend display shows both the proportional change of output based on change of SPE (Step change), followed by proportional change of output based on change of PVE (Exponential approach to a steady value).

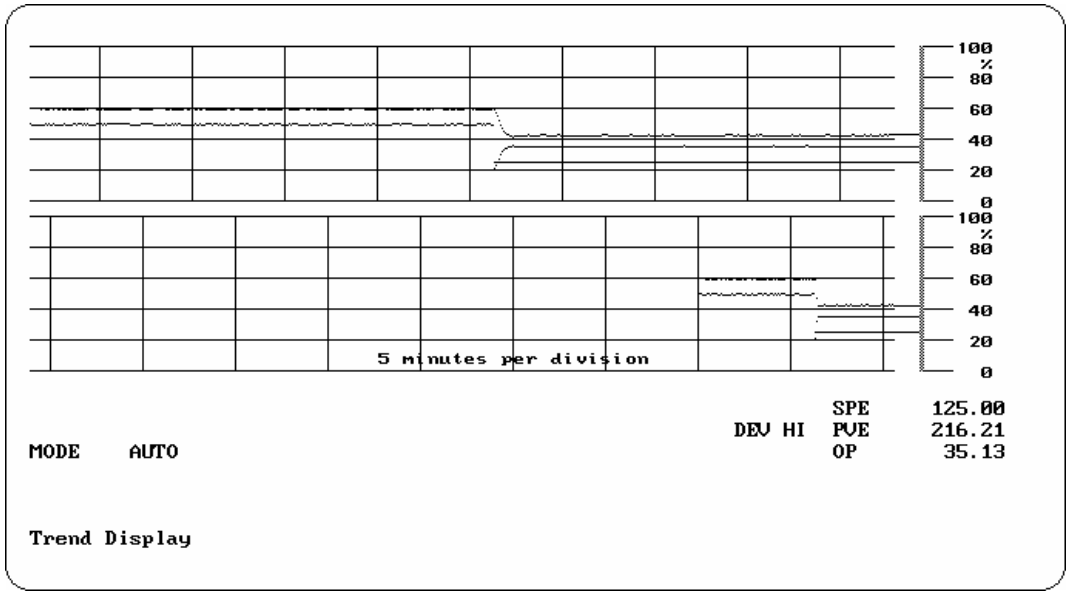


Figure 2.3
Proportional Control Trend with Offset

2.4 Operation

Repeat the above exercise with different values of Gain.

2.5 Conclusion

With increasing values of Gain, we obtain smaller values for OFFSET. This shows, that it may be desirable to use high values for Gain to minimize the Offset. If a control loop has a tendency to be unstable, stability problems put limits on the increase of Gain. Even if no stability problem exists, the value of Gain should be kept as low as possible. This avoids unnecessary amplification of noise. As a result, we learn that it is practically impossible to reduce OFFSET to zero in an industrial control situation.

Exercise 3

Integral (I) Control - Flow Control

3.1 Objective

This exercise will introduce the Integral Control action of controllers. Special emphasis is given to the task of eliminating the remaining offset term of proportional control. It will also show the slower control action of Integral control, compared with Proportional control. Figure2.1 (See Exercise 2) shows an example for closed loop control.

3.2 Operation

Call up the training application "**Single Flow Loop**". After this exercise has been called up, press F3 to get the Detail Display of the flow controller.

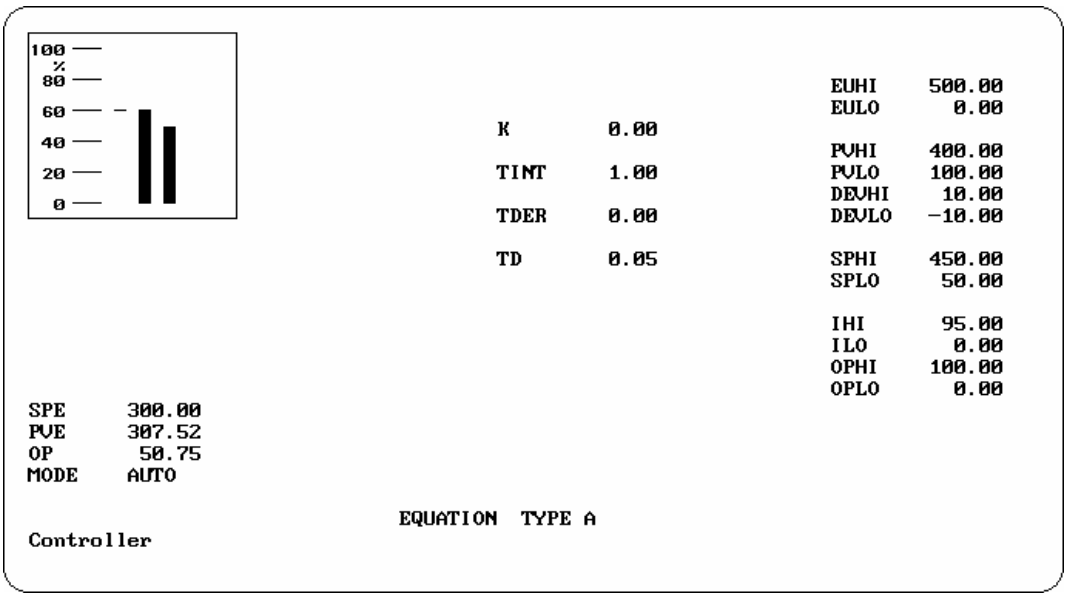


Figure 3.1
Flow Controller Detail Display for I-Control

To prepare the controller for I-Control only, set K to 0, TINT to 1 and TDER to 0. K = 0 causes the controller to switch to Integral control only, using a unit Gain of 1 for Integral only. Gain for Proportional and Derivative control action is 0, as the Detail Display shows (See Figure3.1).

To study I-Control, change SPE from 300 (60% of range) to 125 (25% of range) as shown in Figure3.2. Then change SPE back to 300 (A value of 310 has been used to create Figure3.3).

3.3 Observation

It is observed, with Integral control, that the PVE value settles down exactly at the value of SPE. **No Offset** is left, as would have been with Proportional control only. It can also be observed that the control action does not start with a step, as a result of a step of SPE, as it did with Proportional control. The control action resulting from Integral control action is slow because it has a lagging behavior as shown in Figures3.2 and 3.3.

3.4 Operation

Repeat the above exercise with different values of TINT.

3.5 Conclusion

Integral control will control the PVE towards the SPE precisely, without any OFFSET. The trade-off for this, when compared with Proportional control, is a lagging behavior, which results in slower control. It will be shown later, that Integral control decreases the stability of the loop, if an intrinsic stability problem exists within the control loop. Integral control and its effects on stability are dealt with in chapter 5, *"Tuning of Closed Loop Control"* and exercises 6, 7 and 8.

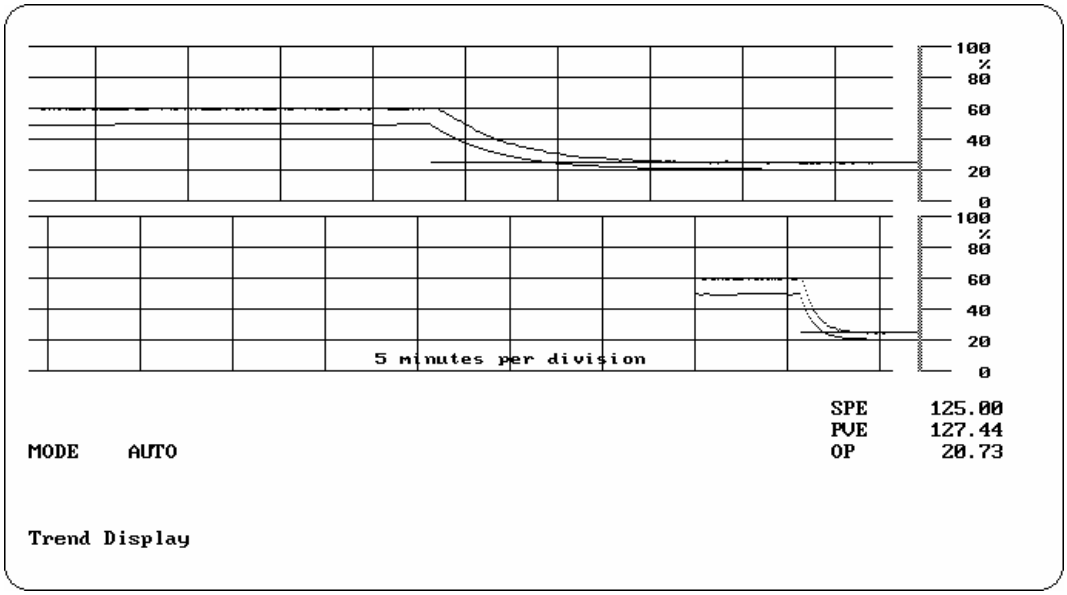


Figure 3.2
Integral Control Trend (SPE 300 to 150)

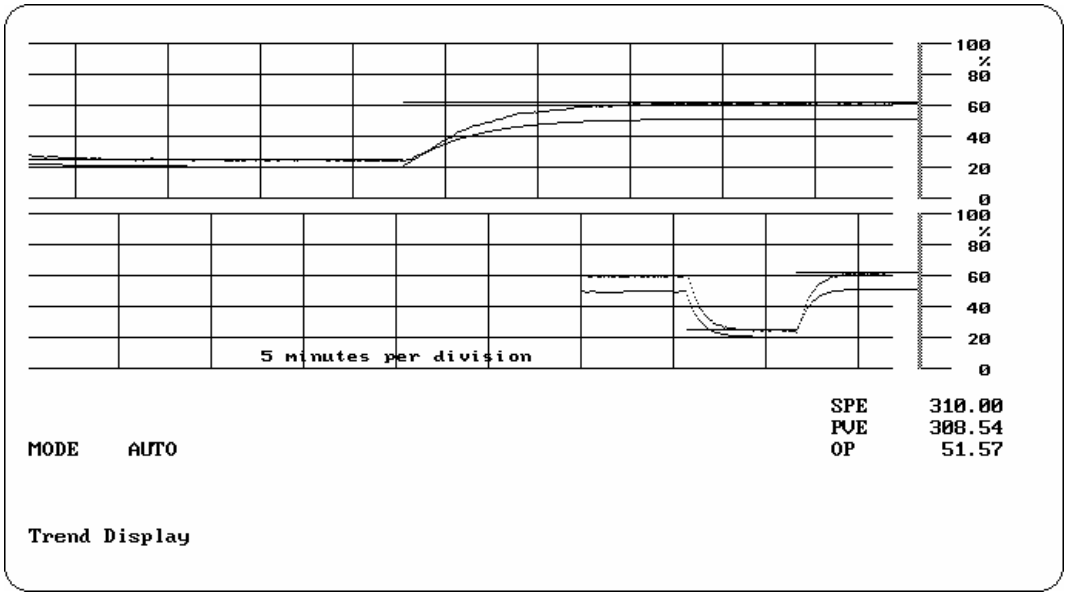


Figure 3.3
Integral Control Trend (SPE 150 to 310)

Exercise 4

Proportional and Integral (PI) Control - Flow Control

4.1 Objective

This exercise will introduce a combined Proportional and Integral Control action of controllers. Special emphasis is given to the elimination of the remaining offset of proportional control without loss of control speed. It will be shown that the combination of Proportional and Integral control maintains the speed of control as it exists with Proportional control only, but without the disadvantage of an OFFSET term. Figure2.1 (See Exercise 2) shows an example for closed loop control

4.2 Operation

Call up the training application "**Single Flow Loop**". After this exercise has been called up, press F3 to get the Detail Display of the flow controller.

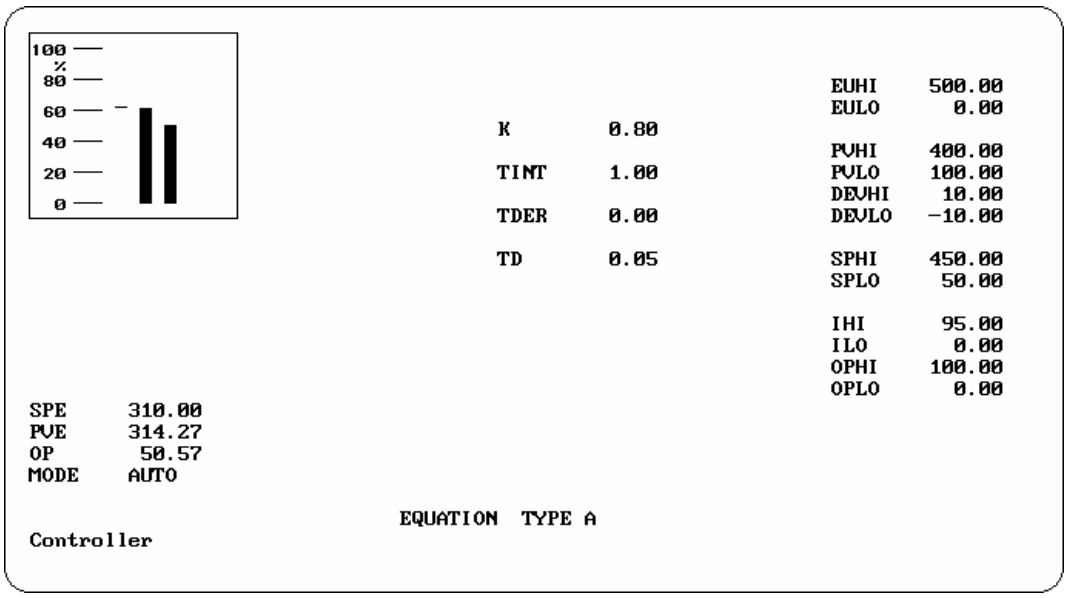


Figure 4.1
Flow Controller Detail Display for PI-Control

To prepare the controller for PI-Control, change K (Gain), TINT (Integral Time Constant) and TDER (Derivative Time Constant). Set K to 0.8, TINT to 1 and TDER to 0.

To study PI-Control, change SPE from approximately 300 (60% of range) to approximately 125 (25% of range) as shown in Figure4.2.

4.3 Observation

It can be seen, that combined Proportional and Integral control moves the PVE value closely towards the value of SPE. **No Offset** is left, as would have been with Proportional control only. There has also been no noticeable loss of speed of control. Close observation of the trend display (See Figure 4.2) first reveals a fast change of PVE, followed by a slow approach towards SPE. The fast change of PVE is the direct result of Proportional control and the slower change of PVE which follows is mainly the result of Integral control.

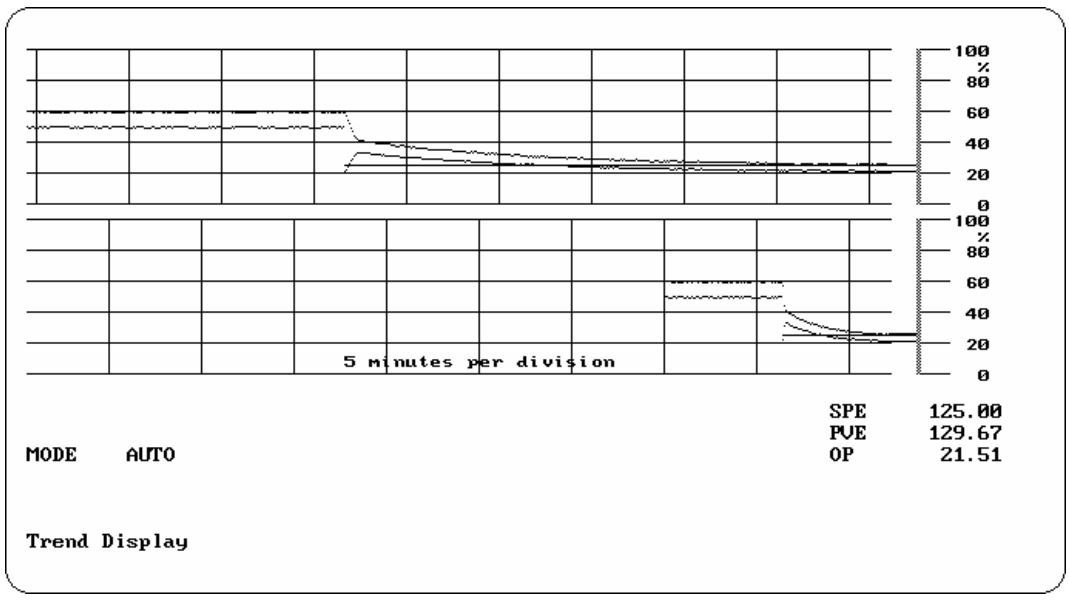


Figure 4.2
Combined Proportional and Integral Control

4.4 Operation

Repeat the above exercise with different values of TINT. Keep K below 1 and use small values for TINT (in our example 0.05 to 0.5), as is done in most practical applications of flow loops.

4.5 Conclusion

Combined Proportional and Integral control turns out to be the best choice of control, if no stability problem exists within the control loop. As a general rule, (with few exceptions) flow control loops have no stability problems. This is the reason why the majority of flow control loops use PI-Control. Stability problems are dealt with in chapter 5, *"Tuning of Closed Loop Control Systems"* and exercises 6, 7 and 8.

Exercise 5

Introduction to Derivative (D) Control

5.1 Objective

This exercise will introduce the Derivative Control action of controllers. Derivative Control action is required only in control loops with stability problems. Derivative control counteracts the instability (oscillation) of control loops. As such it is an additional control action. Derivative Control is always combined to form a PD or PID-Control loop system.

As flow loops generally have no stability problems, a general single loop is used to demonstrate D-Control. Figure5.1 shows a single loop to control temperature. This is an example for a control loop, where a stability problem can be expected. It would create unrealistic responses if we used derivative control in a flow control loop. A general control loop is shown in Figure5.2.

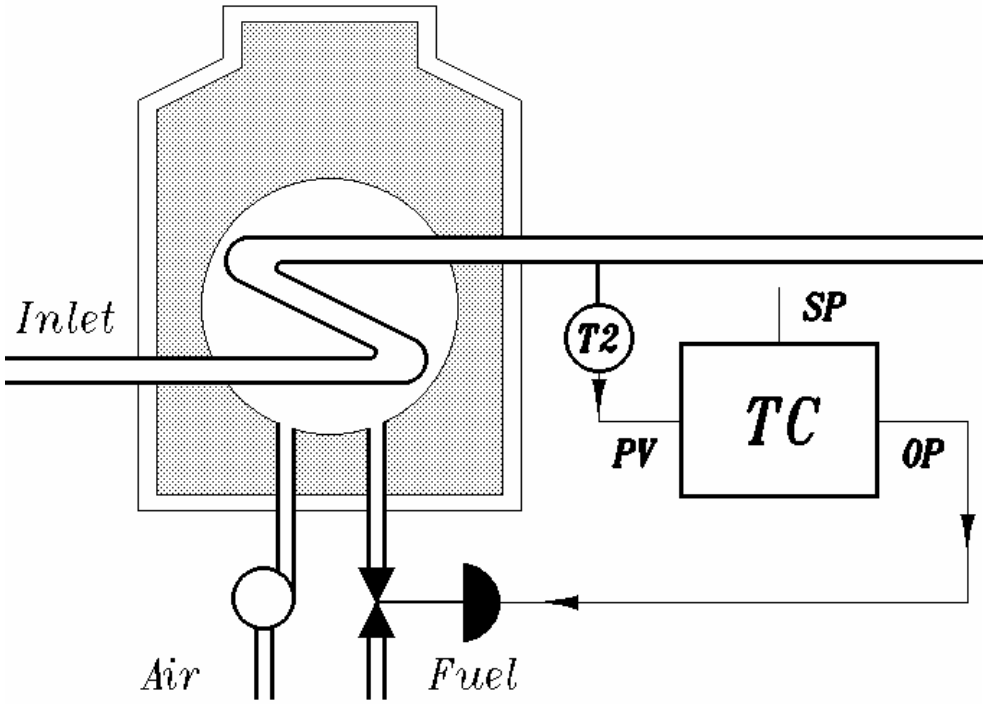


Figure 5.1
Feed Heater Single Loop Control

5.2 Operation

Call up the training application "**General Single Loop with Interactive PID (Real Form)**".

To confirm K-DIST is at 0, call up display F8 (Auxiliary Display). If necessary, change the Gain for the disturbance to 0 (K-DIST = 0). It is important to be aware that some process noise still exists (K-NOISE = 0.1). See Figure5.3.

To study D-Control alone, call up the Detail Display F3 and change K, TINT and TDER to the values shown in Figure 5.4.

In order to observe D-Control, change SPE from approximately 1000 (50% of range) to approximately 1200 (60% of range). Figure5.5 shows an example where SPE has been changed from 1000 to 1220.

5.3 Observation

It is observed that Derivative control calculates a controller OP value based on the Rate of Change of PVE-SPE. As is explained in greater detail in the chapter "Digital Control Principles", D-Control of an "Interactive PID Controller" is in actual fact a PD-Control action with a Low-Pass-Filter (See Figure II.14). Therefore, the initial OP change as a result of the step change of SPE is not a Unit-Pulse-function (Needle shaped pulse with magnitude); it is in fact a limited step of OP, followed by an exponential decay. The initial step is 8 * K * (PVE-SPE). As a P-component always exists within this control action, PVE settles down with an OFFSET. Exercise10 will demonstrate the different kinds of D-Control in more detail.

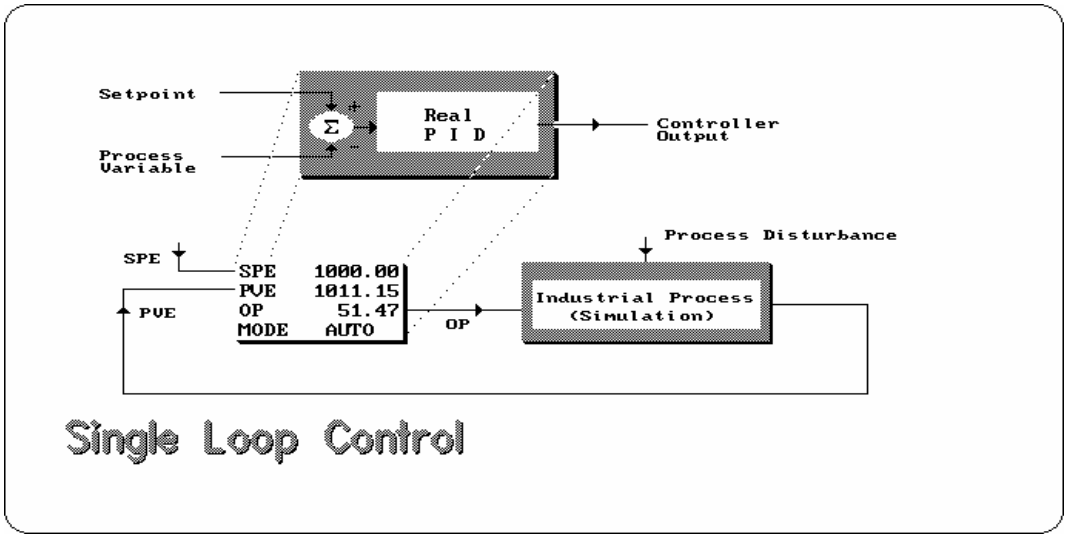


Figure 5.2
Single Loop Control Block Diagram

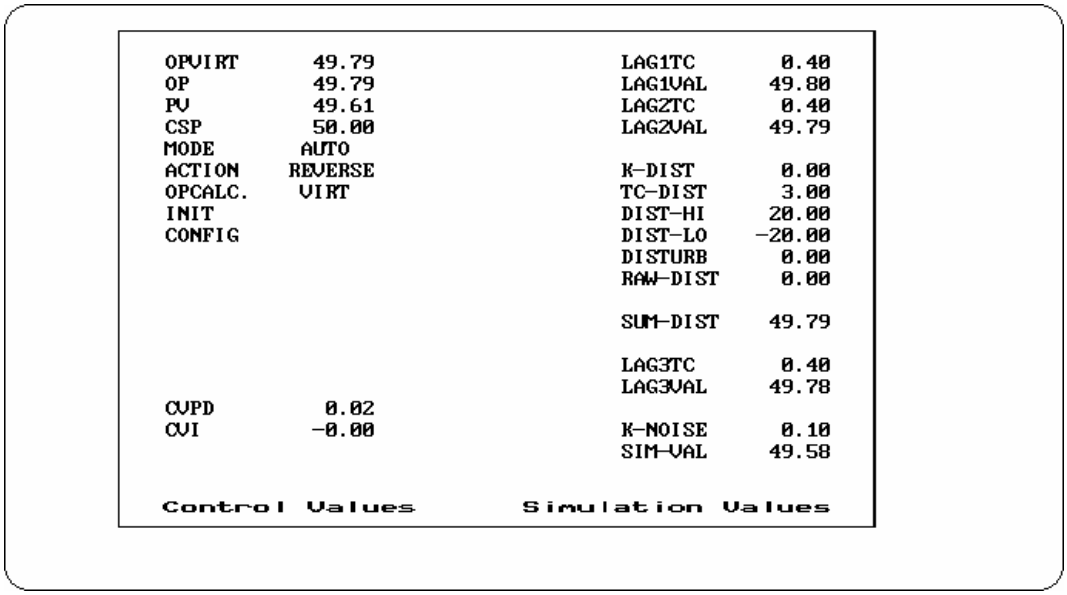


Figure 5.3
Auxiliary Display (F8)

5.4 Operation

Repeat the above exercise with different values of TDER and K. Observe the clipping of OP at the OPHI and OPLO limits, if large values of K are used.

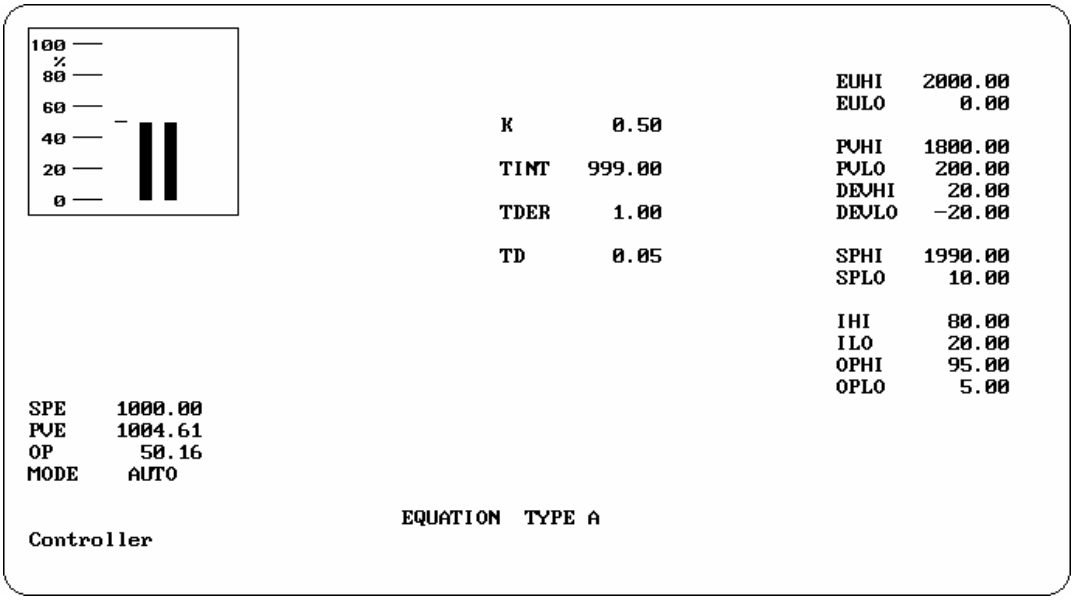


Figure 5.4
Controller Detail Display for D-Control

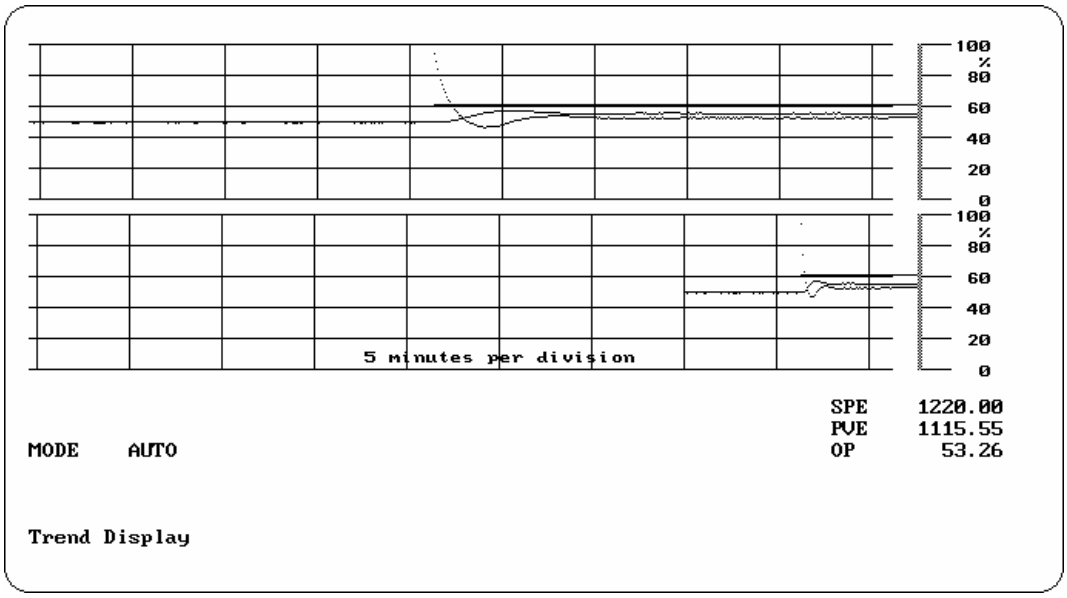


Figure 5.5
Trend Display for D-Control

5.5 Conclusion

Derivative Control is based on the rate of change of PVE-SPE and is not designed to bring the value of PVE to the value of SPE. The sole purpose of D-Control is to stabilize an intrinsically unstable control loop.

Exercise 6

Practical Introduction into Stability Aspects

6.1 Objective

The objective of this exercise is to demonstrate the direct relationship between process lag and stability, and to make students aware of the great difference between noise and stability in practice.

6.2 Operation

Call up the training application ***"General Single Loop with Interactive PID (Real Form)"***.

In order to see the stability problem of closed loops in practice and without disturbances, confirm that K-DIST is 0. Call up Auxiliary Display F8 and change the Gain for disturbance to 0 (K-DIST = 0) and the Gain for Noise to 0 as well (K-NOISE = 0). Then the Auxiliary Display should be as shown in Figure 6.1.

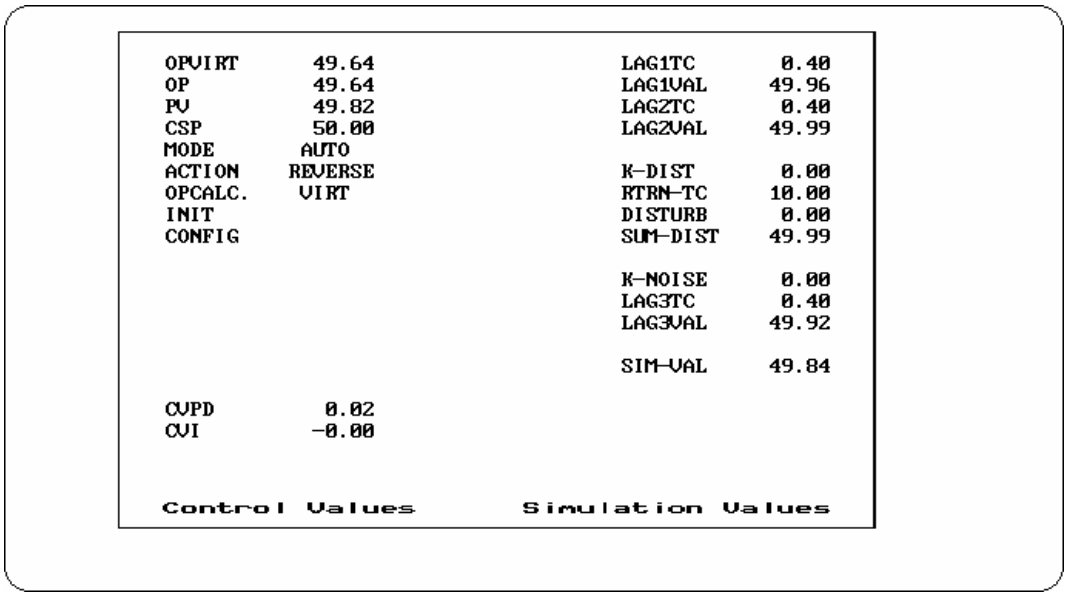


Figure 6.1
Auxiliary Display Showing K-DIST = 0 and K-NOISE = 0

Call up Detail Display F3 as well, and make the changes K = 3.5, TINT = 999 and TDER = 0 as shown in Figure 6.2. The values shown in the Detail Display make it possible to create a continuous oscillation, purely based on the lag of the process. With these values, no integral and no derivative control action takes place and the controller will neither increase nor decrease the phase shift within the control loop. Therefore, the existing phase shift has been caused by the process only.

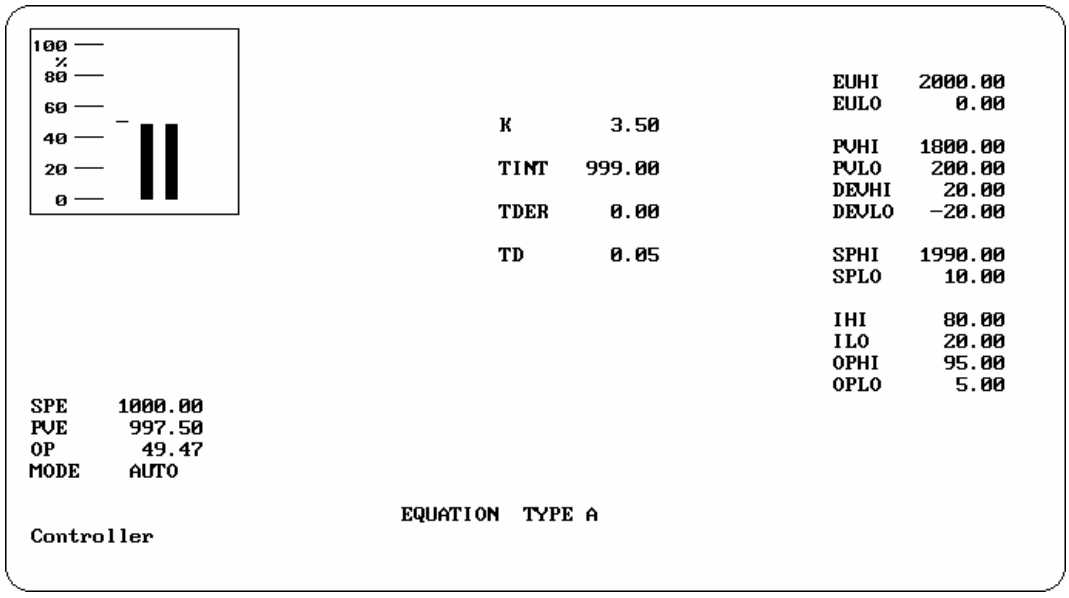


Figure 6.2
Detail Display

Change SPE from 1000 to 1100 to start continuous oscillation. See Figure 6.3.

6.3 Observation

In this situation, the controller performs P-Control only. As shown in the chapter "Stability and Control Modes of Closed Loops", continuous oscillation exists if Loop Gain is 1 and Loop Phase Shift is 180 . Since our controller Gain K is 3.5 we can calculate a process Gain of 0.3.

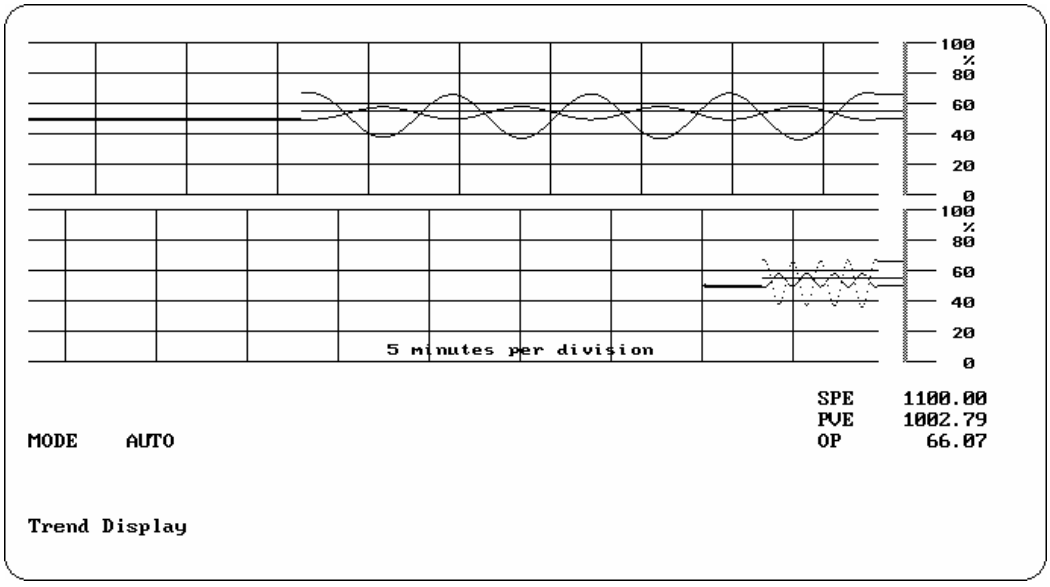


Figure 6.3
Continuous Oscillation

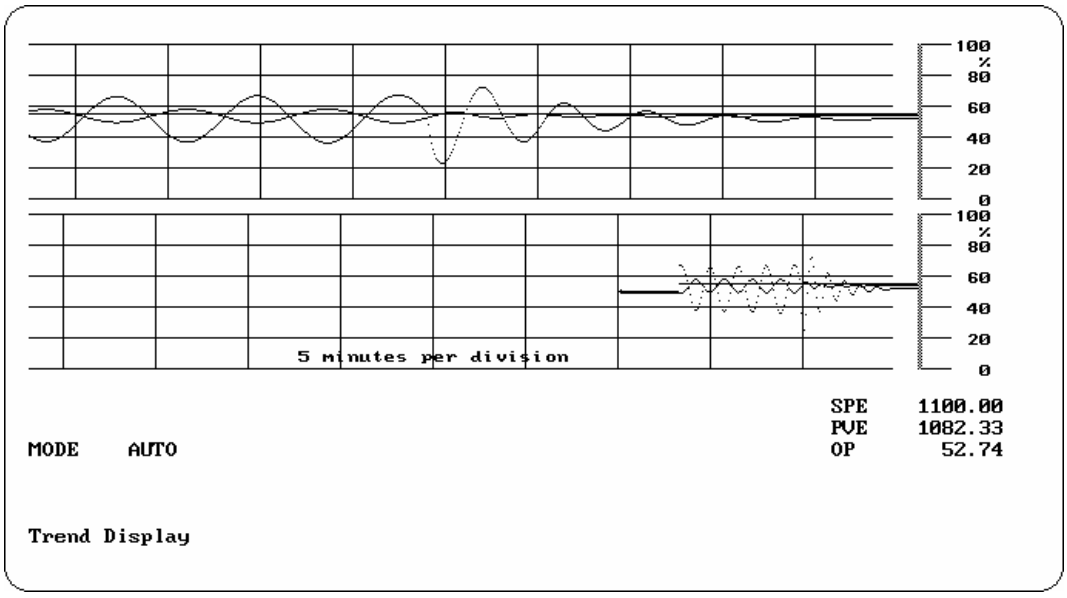


Figure 6.4
Transition from P-Control to PD-Control

6.4 Operation

In the following operation, we will add derivative control to the existing situation of continuous oscillation, in order to find out what effect it will have on stability. Change TDER in the Detail Display to 0.5 (TDER = 0.5). This will result in a Trend Display (F4) as shown in Figure6.4.

6.5 Observation

The left side of the Trend Display (Figure 6.4) shows continuous oscillation with P-Control only. The right side of the Trend Display shows the effect of added D-Control. The trend shows obviously faster control combined with a suppression of the oscillation. This means that **D-Control has a stabilizing effect.**

6.6 Operation

It is vital for every Process Control Engineer to have experience in the effects of Process Noise and D-Control. Through the following operations this will be experienced. First change K-NOISE in display F8 to 0.5. (The value of K-NOISE is different with different processes.) Now repeat the same operation as before (K =3.5, TDER + 0, TINT + 999), in order to create continuous oscillations (but now superimposed by noise). Then add D-Control (TDER + 0.5). Figure 6.5 shows the Trend Display resulting from this exercise.

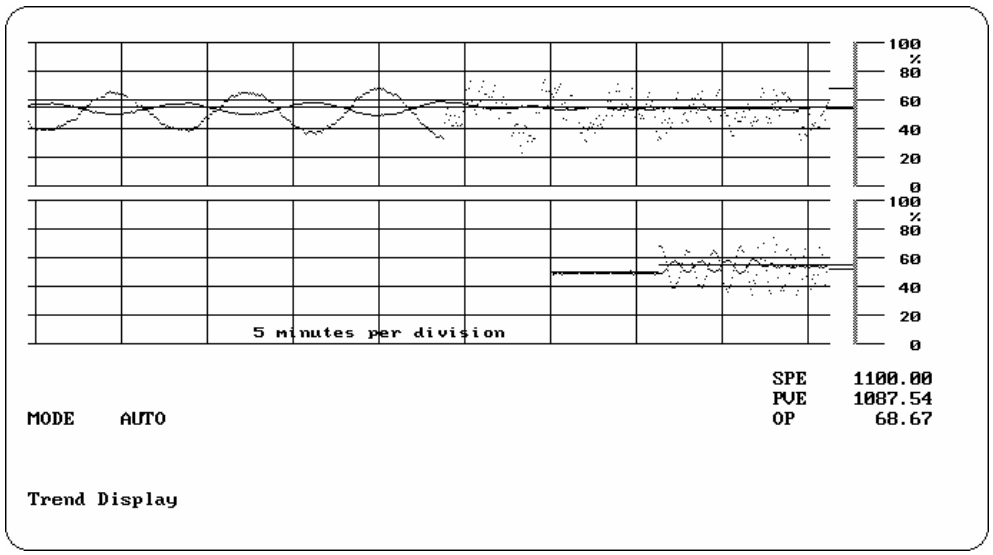


Figure 6.5
Transition from P-Control to PD-Control of a Noisy Process

6.7 Observation

It is clearly visible, that D-Control multiplies noise.

6.8 Operation

Now repeat the exercise as above, but add I-Control, (not D-Control as before), in a continuous oscillating situation created by P-Control. The student adds I-Control in order to change from P-Control to PI-Control. Use TINT = 0.5 to obtain a Trend Display as shown in Figure 6.6. Then start again with continuous oscillation, created by P-Control only and change to I-Control only.

To do this, change K from 3.5 to 0 and TINT from 999 to 0.5. The change from P-Control to I-Control is shown in Figure 6.7. The continuation of this trend is shown in Figures 6.8 and 6.9.

6.9 Observation

Figures 6.6 to 6.9 show, that integral action has a destabilizing effect, whether it is combined PI-Control or I-Control only. In both cases we observe an increase of magnitude in oscillation. In addition we observe, that I-Control ignores noise. We can see in Figure Ex.6.6, that the change from P-Control to PI-Control has no effect on noise. Noise is not different in P-Control and PI-Control. Noise is almost totally suppressed when I-Control is used only (see Figures6.7 to 6.9).

6.10 Conclusion

This exercise brings us to the following conclusions:

- D-Control has a stabilizing effect, whereas I-Control has a destabilizing effect. The only purpose of I-Control is to eliminate the offset of PonlyControl.

- The trade-off is the destabilizing effect. This may be compensated by adding D-Control. This will result in the use of PID-Control for control loops with stability problems.

- D-Control has to be treated very carefully from a noise point of view. It is important to point out that D-Control should never be used without prior filtering of the PVE. Such a filter has to reduce noise without adding a significant Lag to the loop. The student will find in the Detail Display F3 a variable TD. TD is the filter time constant for PV filtering.

Note:
Never use D-Control without PV filtering (TD) if any process noise exists.

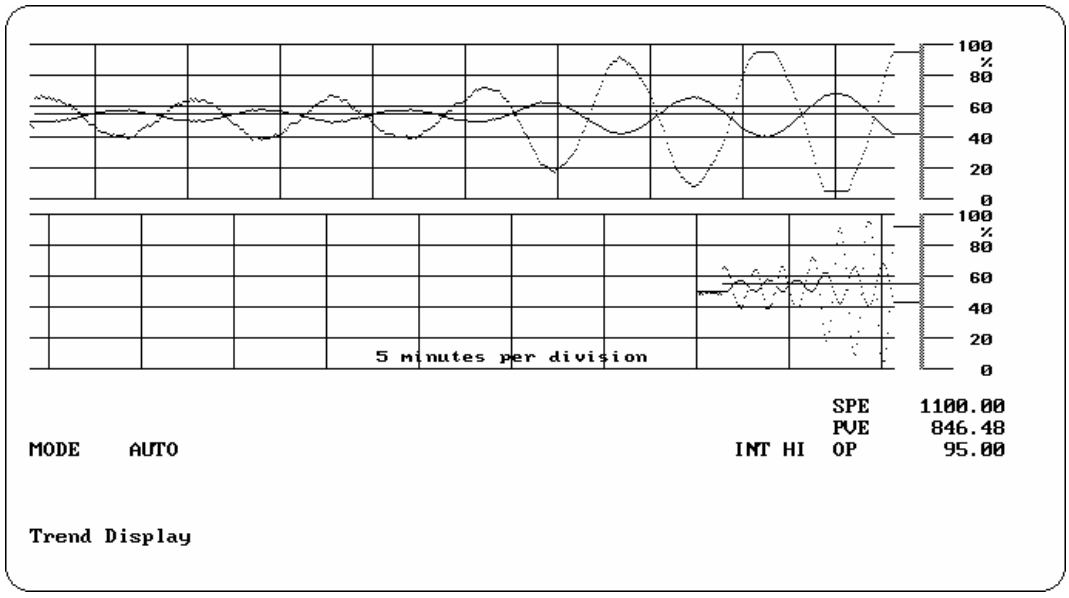


Figure 6.6
Transition from P-Control to PI-Control of a Noisy Process

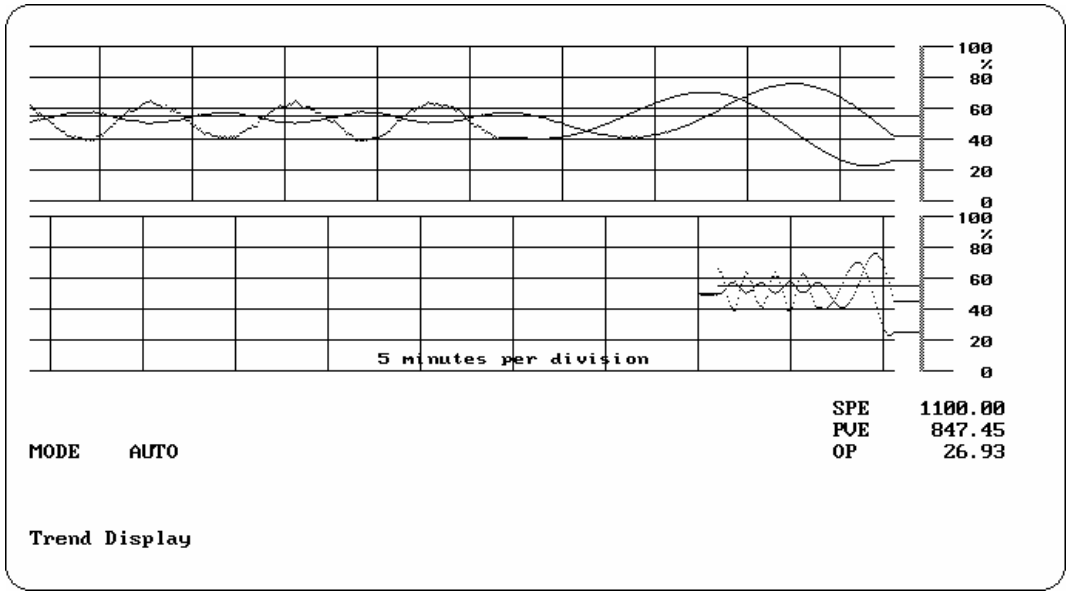


Figure 6.7
Transition from P-Control to I-only-Control of a Noisy Process

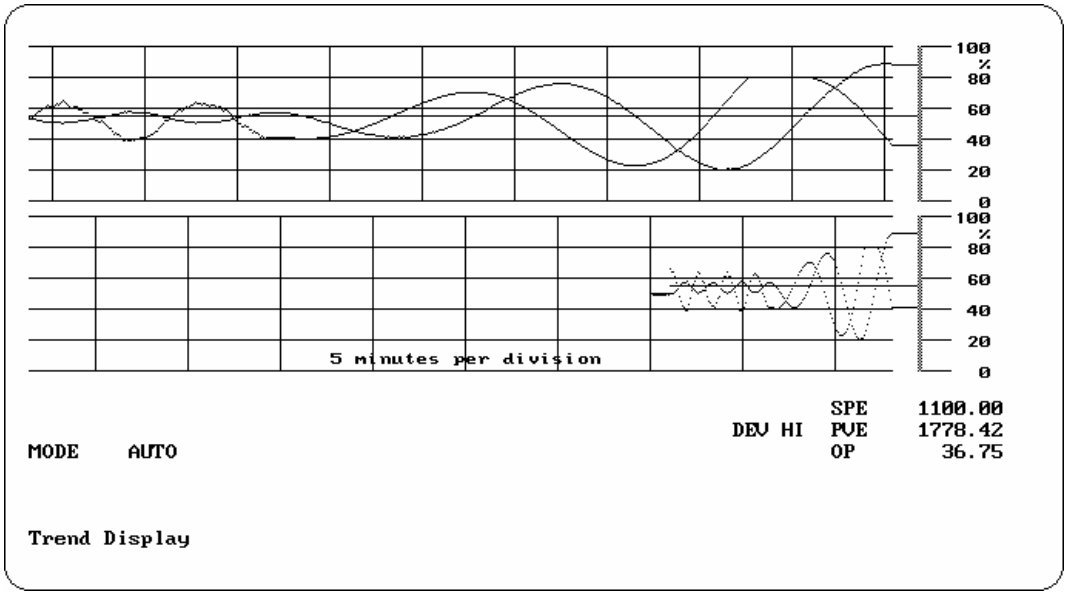


Figure 6.8
Transition from P-Control to I-only-Control of a Noisy Process
(cont. of Figure 6.7)

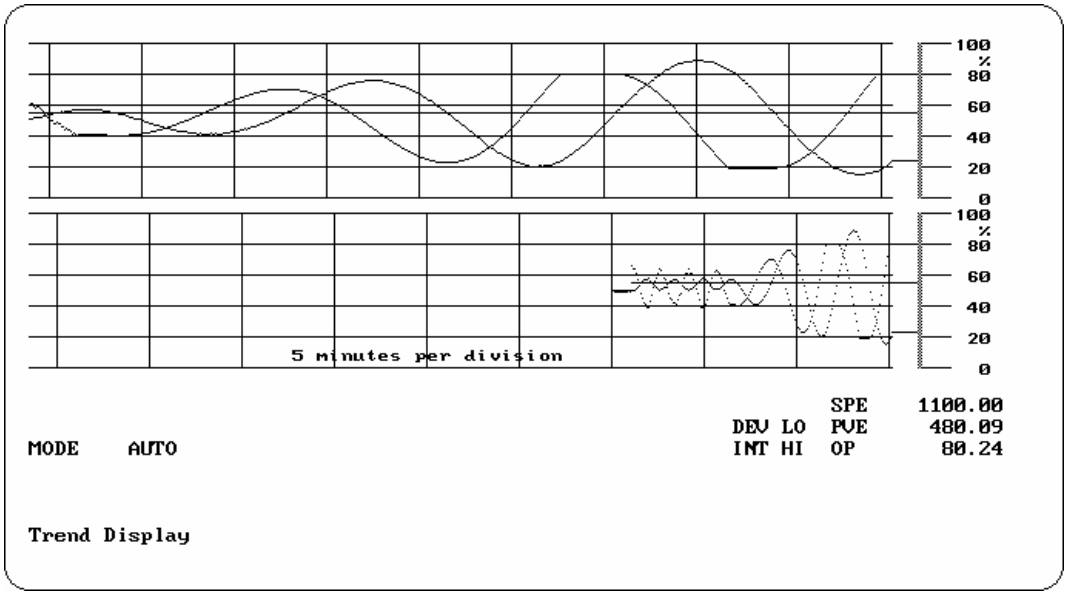


Figure 6.9
Transition from P-Control to I-only-Control of a Noisy Process
(cont. from Figure 6.8)

Exercise 7

Open Loop Method -Tuning Exercise

7.1 Objective

This exercise will give some practical experience in the Reaction Curve Method of tuning. Figure 7.1 shows the diagram of a typical control loop. The steps for tuning are explained in chapter 5, "Tuning of Closed Loop Control Systems".

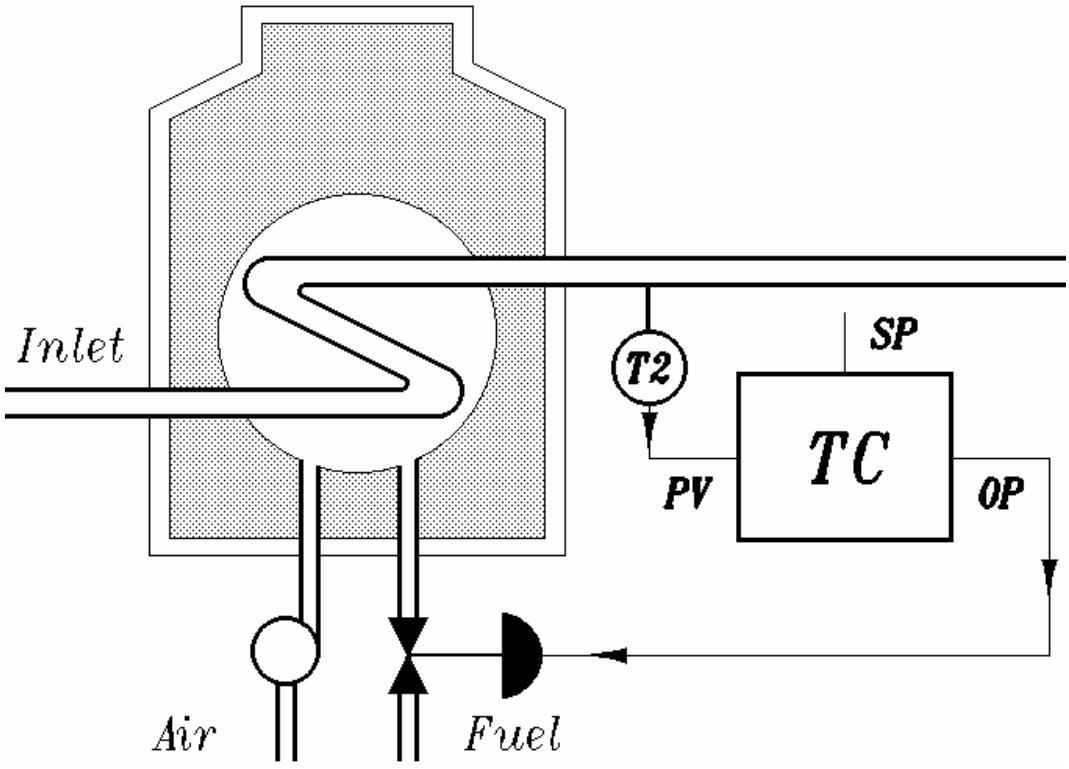


Figure 7.1
Single Loop Control

7.2 Operation

Call up the training application "**General Single Loop with Interactive PID (Real Form)**".

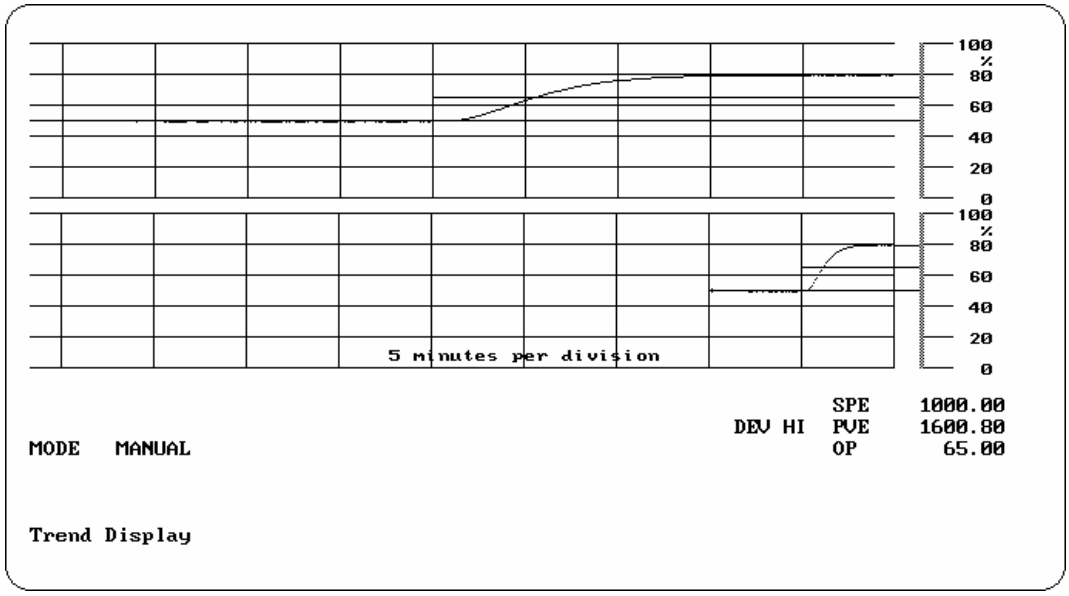


Figure 7.2
Reaction Curve Trend Display for Open Loop Tuning

Call up the Detail Display F3 and change MODE to MANUAL. Make a good judgment about the noise observed on the PV and set the digital noise filter (TD) accordingly. A good value of TD will result in a still noticeable but very small magnitude of noise on PV. The noise filter has to be set up before tuning in order to have it included in the result of tuning. From a practical point of view, the noise filter becomes part of the process as far as loop tuning is concerned. In this exercise, a good value for TD will be between 0.05 and 0.1.

Make a small change of OP, to find out how sensitive the process is in its response. Then wait until the process is steady again (until PV is fairly constant). Now make a change of OP with a magnitude which ensures that the PV stays within its Alarm Limits PVHI and PVLO (Ignore the violation of DEVIATION limits in this exercise).

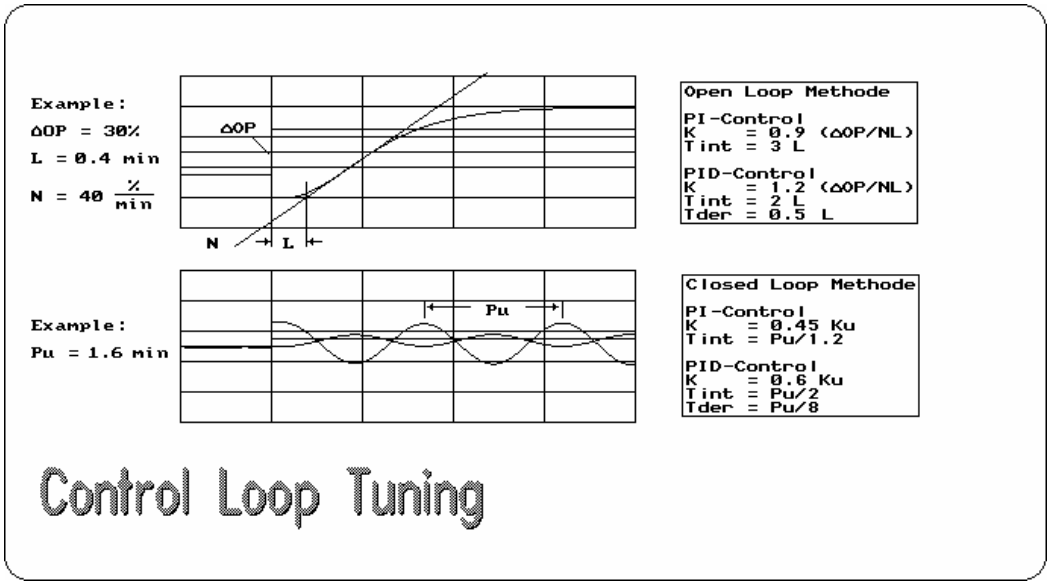


Figure 7.3
Tuning Formula Display

To obtain a Process Reaction Curve as shown in Figure7.2, set MODE to MANUAL, then OP to 50 and wait until the PVE is steady at approximately 1000 (50% of range). Then make a change of the OP from 50 to 65.

7.3 Observation

Observe the reaction curve and tune the controller as explained in chapter 5, *"Tuning of Closed Loop Control"*. Call up display F6 to get the summary of the major tuning formulas on screen as shown in Figure 7.3.

The values obtained from the reaction curve are:

ΔOP = 15% (from 50% to 65%)
N = 20% / minute (PVE range from 0 to 2000)
L = 0.4 minutes

Using the formulas shown in Display F6, we obtain the following tuning constants for PID-Control:

K = 2.2
TINT = 0.8
TDER = 0.2

After setting the tuning constants to their calculated values, the Detail Display F3 should look as shown in Figure 7.4.

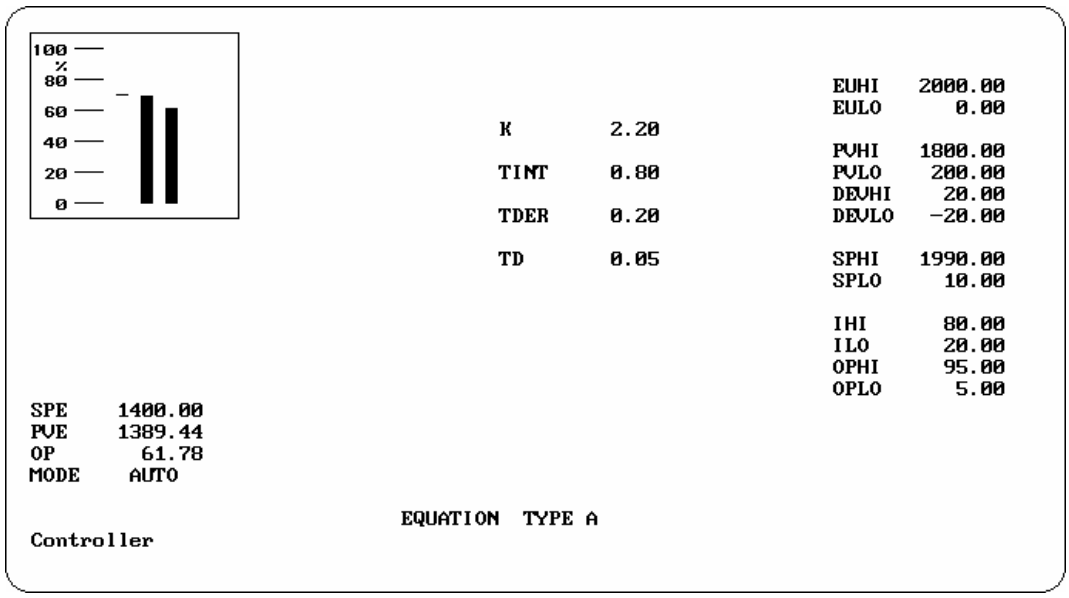


Figure 7.4
Detail Display with Tuning Constants

7.4 Confirmation of Proper Tuning

Change Mode to Automatic and make a change of the SP. Observe the process settling down (Observe PV). Use the trend display as shown in Figure 7.5 for observation. Most parameters can be operated from the trend display as well. As a rule of thumb, the settling down should take place with quarter damping (1/4 decay). That means two successive maximum values of a damped oscillation should have a magnitude ratio of 1 to 4.

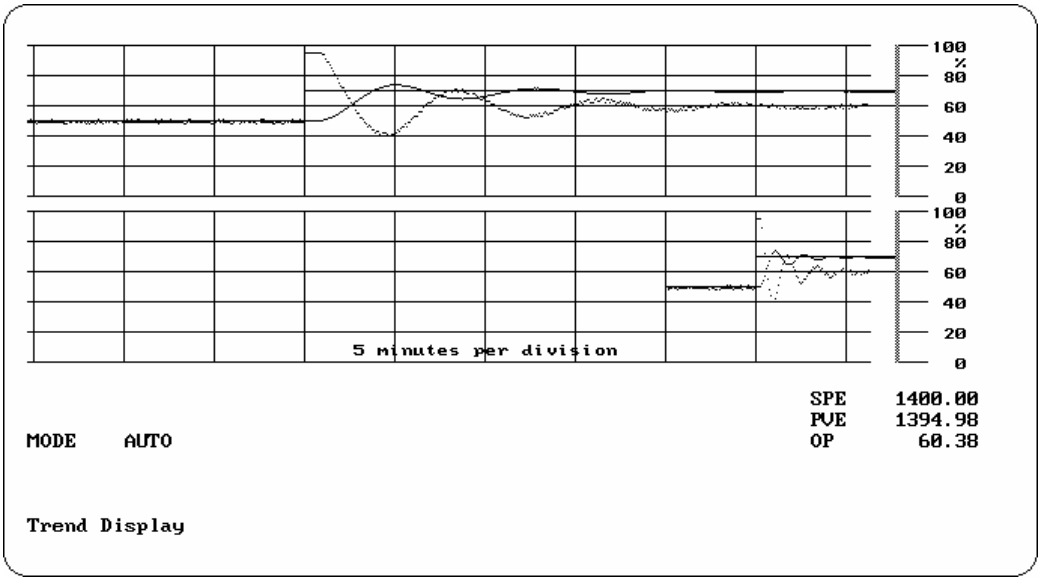


Figure 7.5
Trend Display of Tuned Loop

7.5 Fine Tuning

Based on process knowledge (Noise on PV etc), experience and precise knowledge of the control actions of each control mode (P-Control, I-Control and D-Control) fine tuning of the process should be performed now. Only minor variations to the tuning constants should be made.

- **TD-Fine Tuning**

  Observe the noise of the control action (OP). Use your judgment as to how much to increase or decrease TD. The judgment has to be based mainly on the effect noise in the OP signal has on the manipulated variable (Wear and tear of a valve for instance).

  An increase of TD may require an increase of T2 (Derivative Time Constant) to compensate for the lagging phase shift of the noise filter.

- **K-Fine Tuning**

  Since K is the Gain for all control modes (PID), reducing gain reduces the effect of all control actions equally and vice versa. A reduction of K reduces speed of control and increases stability and vice versa.

- **T2-Fine Tuning**

  After tuning, (using the Ziegler Nichols formulas), the damped oscillation of the process after a change of SP takes place equally around the new SP. Increasing the Derivative Time Constant (T2) will bend the baseline towards the old value of SP and therefore reduce the overshoot. In addition stability in general will be increased. It has to be noted, that the area of error increases as well with increased stability. The reason for this is the slower approach of PV towards the SP.

- **T1-Fine Tuning**

  In most cases Integral tuning should be corrected only in relatively stable loops like flow loops. T1-Fine Tuning should be done with close observation using the trend display. Decreasing T1 increases instability and vice versa.

Exercise 8

Closed Loop Method - Tuning Exercise

8.1 Objective

This exercise will give some practical experience in the Closed Loop Tuning Method. Figure 8.1 shows the diagram of a typical control loop. The steps for tuning are explained in the chapter *"Tuning of Closed Loop Control Systems"*.

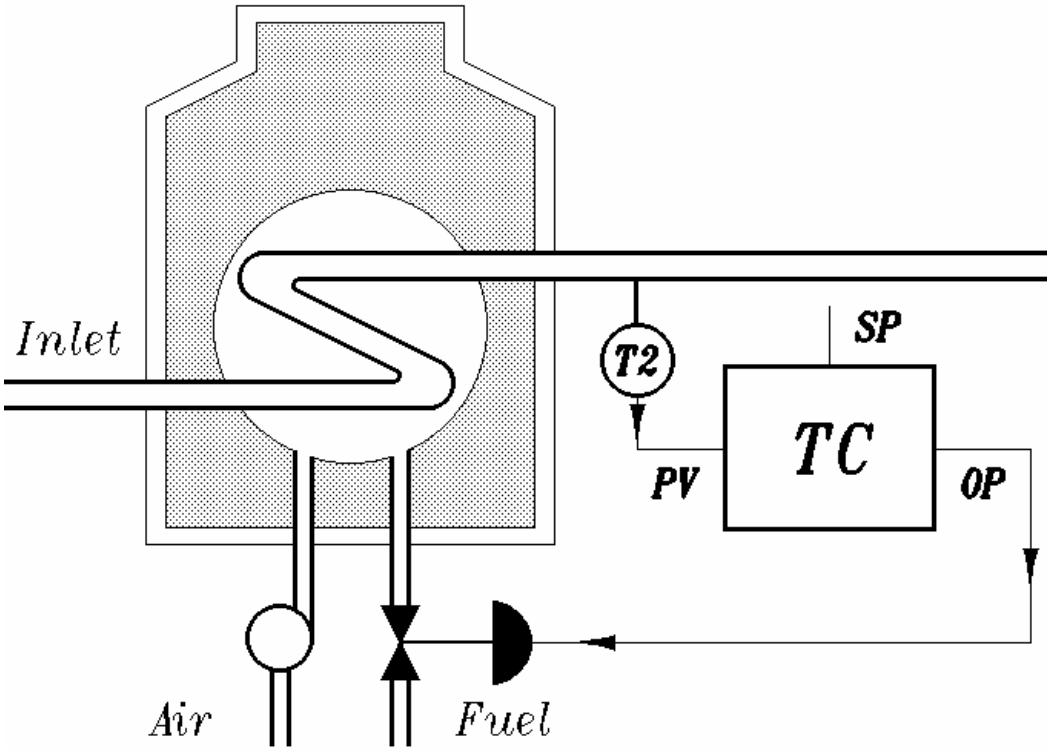


Figure 8.1
Single Loop Control

8.2 Operation

Call up the training application "**General Single Loop with Interactive PID (Real Form)**".

Then, call up the Detail Display F3 and change MODE to MANUAL. Set the OP to 50% and wait for the process to settle down. Make a good judgment about noise observed on PV and set the digital noise filter (TD) accordingly. A good value of TD will result in a still noticeable but very small magnitude of noise in the PV term. The noise filter has to be set up before tuning in order to have it included in the results of the tuning exercise. Practically, the noise filter becomes part of the process as far as loop tuning is concerned.

For the Closed Loop Tuning Method, we have to change the Tuning constants in such a way as to obtain P-Control only. Change TINT to 999, TDER to 0 and K to a relatively low value of 0.5. Then, change MODE back to AUTO and make a small change of SPE in order to find out how sensitive the process is in responding. We have to find a value for K so that the closed loop will oscillate with constant magnitude. If the oscillation settles down, K has to be increased and if the oscillation increases in magnitude, K is too large. Having found the value of gain for constant oscillation, we name this particular value of gain **Ku** (Gain of the Ultimate Frequency). The time period of oscillation we call **Pu** (Period of the Ultimate Frequency).

To obtain continuous oscillation, as shown in Figure 8.2, set MODE to AUTO after the loop has settled down. Then change SPE from 1000 to 1100.

8.3 Observation

Observe continuous oscillation and tune the controller as explained in the chapter "Tuning of Closed Loop Control". Call up display F6 to get the summary of the major tuning formulas on screen as shown in Figure 8.3.

The values obtained from the continuous oscillation Trend Display are:

Ku = 3.5
Pu = 1.6 minutes

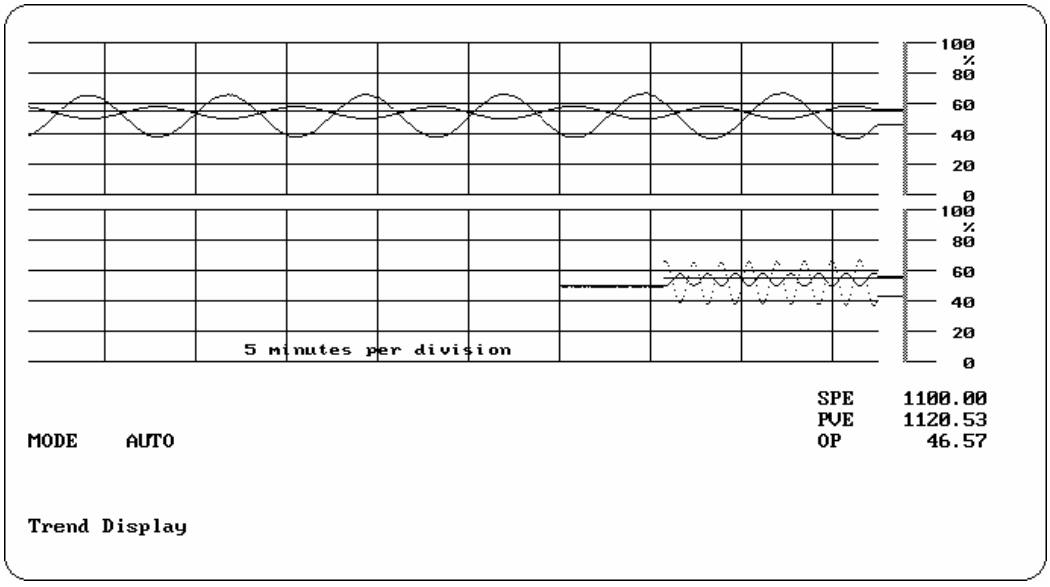


Figure 8.2
Continuous Oscillation Trend Display for Closed Loop Tuning

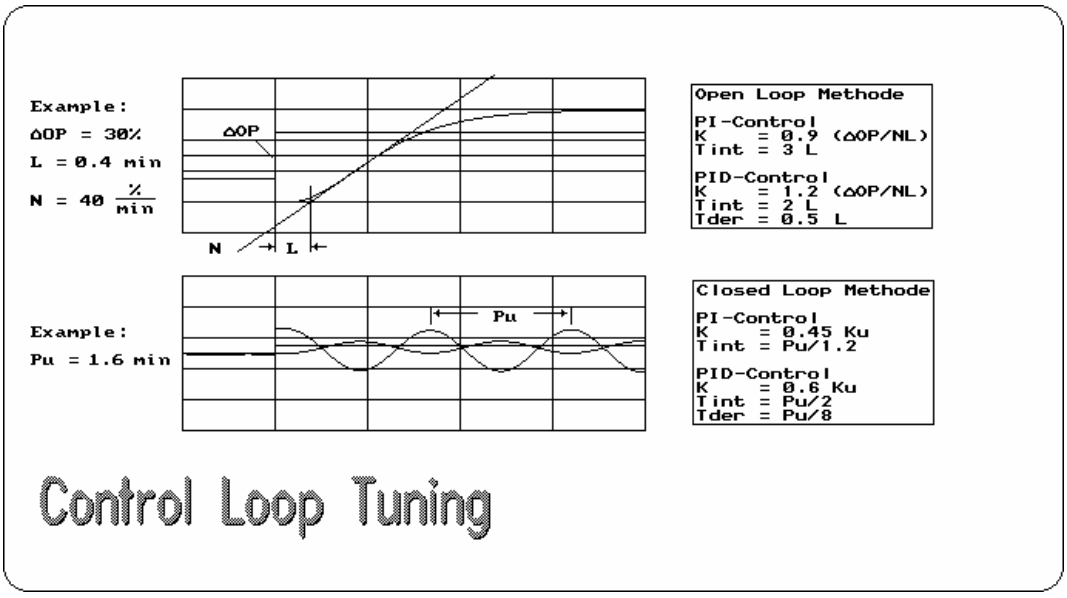


Figure 8.3
Tuning Formula Display

Using the formulas shown in Display F6, we obtain the following tuning constants for PID-Control:

K = 2.2
TINT = 0.8
TDER = 0.2

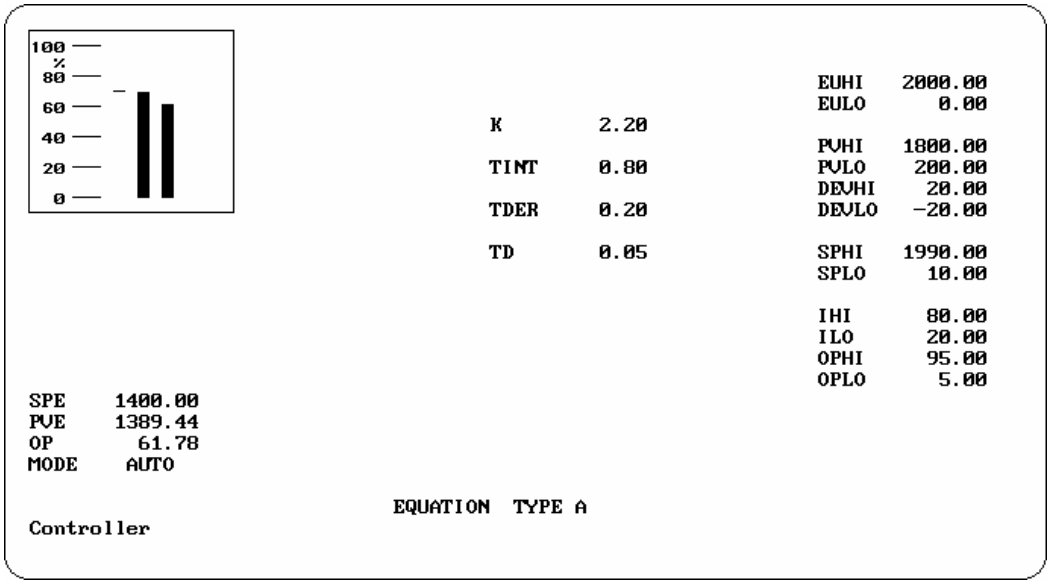


Figure 8.4
Detail Display with Tuning Constants

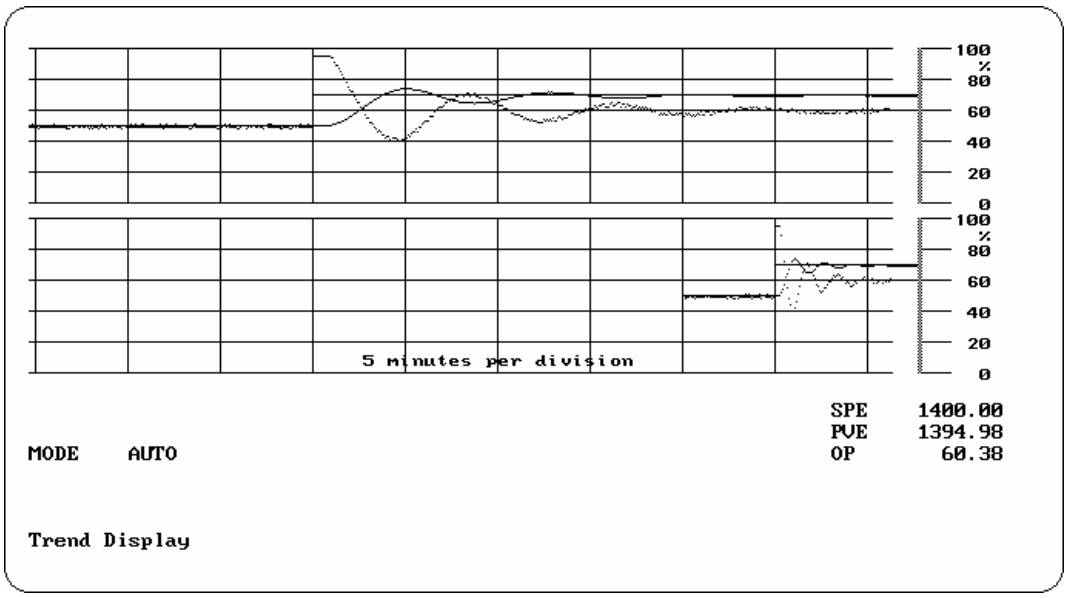


Figure 8.5
Trend Display of Tuned Loop

After setting the tuning constants to their calculated values, the Detail Display F3 should look as shown in Figure 8.4.

8.4 Confirmation of Good Tuning

Wait for the process to settle down and then make a change of SP. Observe the process settling down (Observe PV). Use the trend display as shown in Figure 8.5 for observation. All parameters displayed in the tuning display can be operated from there. As a rule of thumb, the settling down should take place with quarter damping. This means that two successive maximum values of a damped oscillation should have a magnitude ratio of 1 to 4.

8.5 Fine Tuning

Fine Tuning is done in the same way as shown in exercise 7.

Exercise 9

Saturation and Non Saturation Outputs Limits

9.1 Objective

This exercise will introduce the student to the two major types of OP-Limit-Calculations. The one kind, with OP-Limits, allows a saturation value of the OP, based on P-Control and D-Control, and another kind, does not allow OP values to go into saturation at all.

9.2 With Saturation of the OP

The controller calculates a **VIRTUAL OP** value independent of any OP-Limit. These may be values far above 100% or far below 0%. Only the real OP, which is the displayed OP value, is limited by OP-Limits. The real OP awaits the return of the virtual OP within OP-Limits. Then, within the range of the OP-Limits, the real OP follows the virtual OP value exactly. Controllers driving field OP normally use this kind of OP-Limit handling.

9.3 With Non Saturation OP-Limit-Calculation

Only real OP values are used for the calculation. If a single calculation results in an OP value beyond OP-Limits, the OP value will be set to the value of the OP-Limit it would have violated. When the controller calculates the OP value next time (in the next SCAN), the **REAL OP** value (OP = OP Limit) is used. The previous calculation, beyond OP-Limits, has been totally forgotten.

9.4 Virtual OP Limit Operation

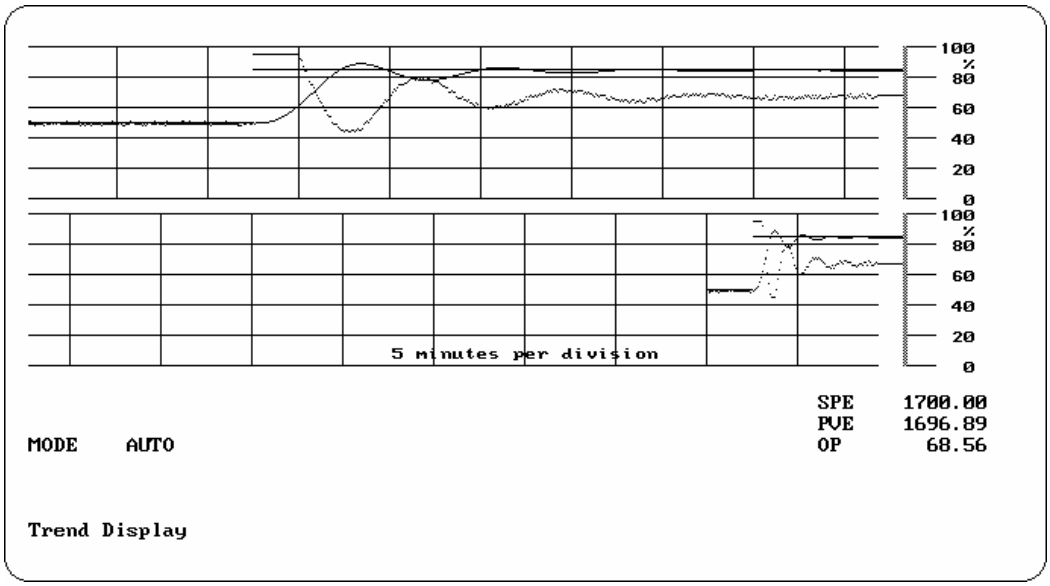


Figure 9.1
Virtual OP-Limit Calculation

9.5 Operation

Call up the training application "**General Single Loop with Interactive PID (Real Form)**". In order to operate with Virtual OP Limit calculation, call display F8 and make sure that the status variable **OPCALC. is set to VIRTUAL**. Then make large changes of SPE (from 1000 to 1700) in order to observe the OP reach saturation. Figure 9.1 shows how the derivative action goes into saturation.

9.6 Real OP Limit Operation

Without Saturation of the OP, there is no difference made between calculated and real OP values. If a calculation of the OP exceeds the OP-Limits, no difference is made between calculated and real output. If the result of an OP calculation would be beyond an OP-Limit, the real OP value is set to the OP-Limit value. No virtual OP value is memorized for the next scan's calculation. Generally, computer resident PID algorithms, not driving field values, use this kind of OP-Limit handling. Before using this kind of OP-Limit Calculation for this exercise, call up display F8 and change the variable **OPCALC**. to **REAL** (See Figure 9.2).

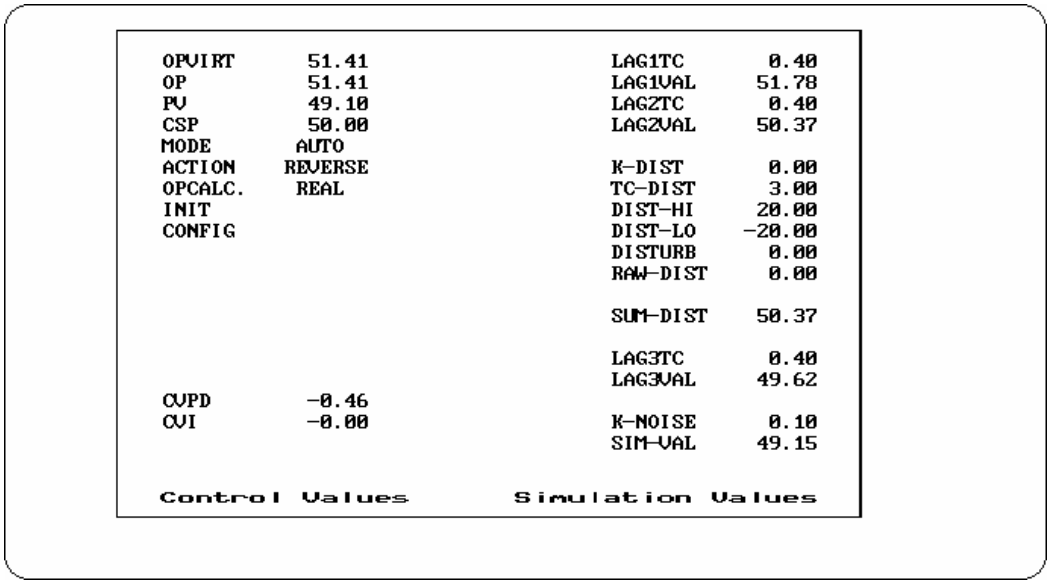


Figure 9.2
Auxiliary Display (F8)

Figure 9.3 shows how the same SP change causes a different derivative action (without saturation of the OP) as shown in Figure 9.1 (with OP saturation).

The Trend Display shown in Figure 9.3 is difficult to understand. The most important thing to observe in this display is a little single dot at 95% at the same time when SPE changed from 1000 to 1700 in a step function. The single isolated dot represents a single OP Calculation limited by OPHI. Already the next calculation of OP includes a decrement of OP so large, that the OP Limit OPLO inhibits OP to be below 5% of range.

Think about the differences and think about possible applications for the two kinds of OP Limit Calculations. **Remember the use of EQ-Type B or C.**

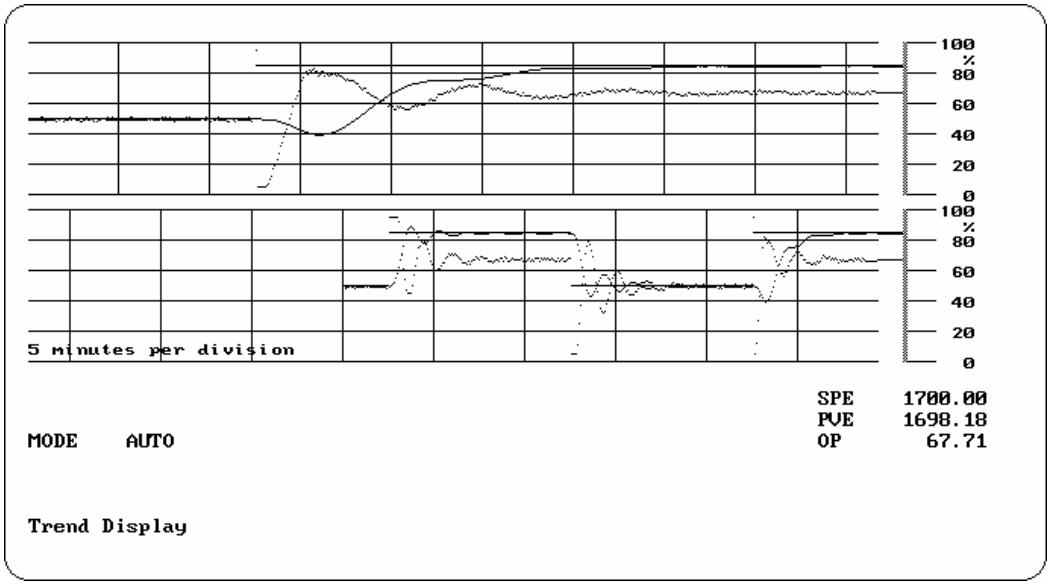


Figure 9.3
REAL OP-Limit Calculation

Exercise 10

Ideal Derivative Action -Ideal PID

10.1 Objective

This exercise will introduce the Non-Interactive Form of the PID Algorithm (**IDEAL PID**). The Ideal PID algorithm makes use of a mathematically true derivative calculation (sT). Generally, Ideal PID control is combined with non-saturation OP-Limit handling. Ideal PID is used for high-level control concepts only (e.g. PID-X in Honeywell equipment). The algorithms reside mainly in supervising computers (as opposed to PLCs, loop controllers or the RTUs of a DCS system). The same SP change can cause a very different (and very confusing) looking derivative control action for either the Ideal or Real PID controller. Compare the derivative control of a Non-interactive PID controller with real OP-Limit calculation (Figure 10.2), to that of an Interactive PID controller with real OP-Limit calculation (Figure 10.3). Detailed knowledge about the practical application is required to make optimum use of this kind of control at the right place. Note that if you are in doubt as to which algorithm to choose, select Real PID control.

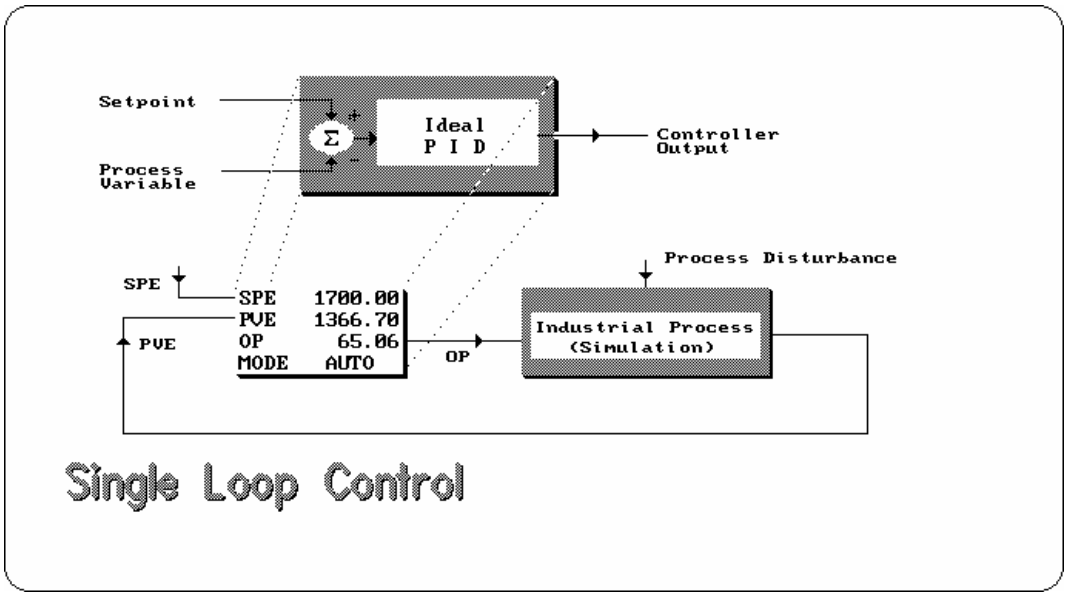


Figure 10.1
Closed Loop Control Block Diagram (IDEAL PID)

10.2 Operation

Call up the training application "**General Single Loop with Non-Interactive PID (Ideal Form)**". Ensure the control loop settles down. The difference in derivative control between Ideal-PID and Real-PID is most noticeable if large values of TDER are required. Whenever loop tuning according to Pessen is indicated, relatively large values of TDER will be used. In our exercise, we make use of the values calculated according to Pessen (K=1.1, TINT=0.8, TDER=0.5 and TD=0.05). Now change SPE from 1000 to 1700 and observe the OP. Figure 10.2 shows the result.

10.3 Observation

The considerable level of noise created by the derivative action of the Non-Interactive PID (Ideal PID) control should be noted.

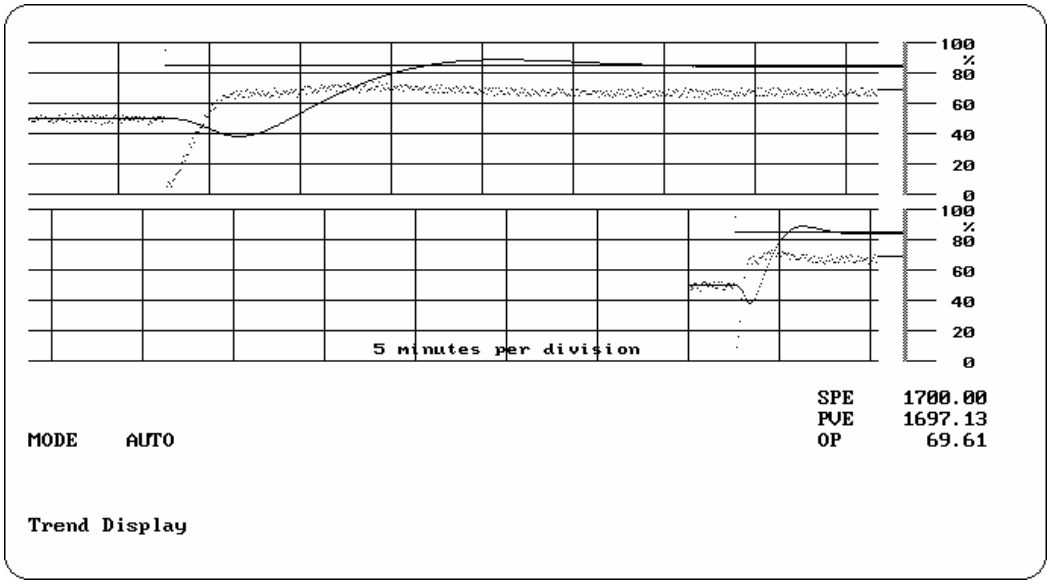


Figure 10.2
IDEAL PID Control with Strong OP-Noise and REAL OP Limit Calculation

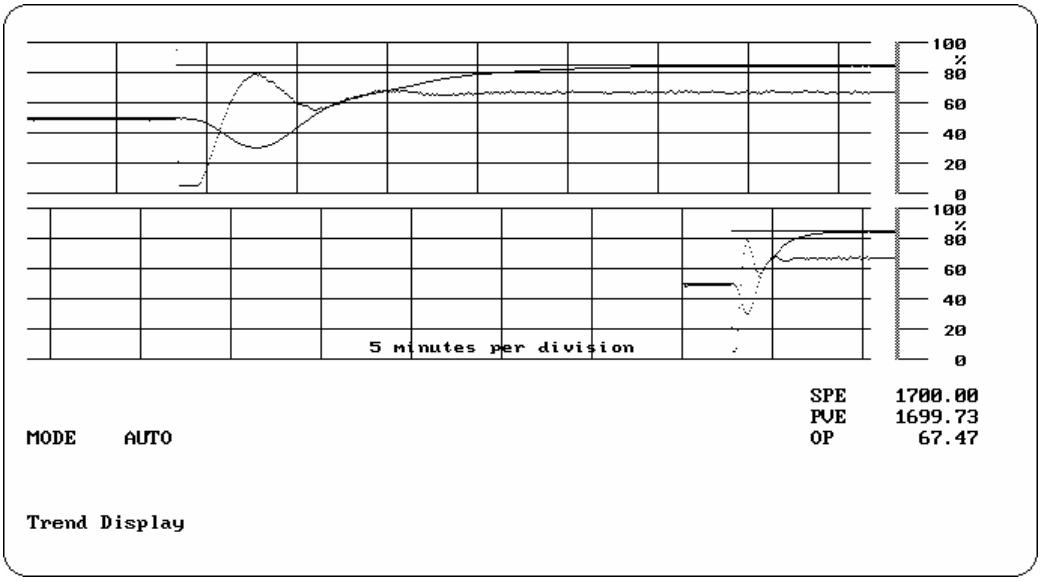


Figure 10.3
REAL PID Control with Some OP-Noise and REAL OP Limit Calculation

10.4 Operation

Call up the training application "**General Single Loop with Interactive PID (Real Form)**". Call up display F8 and change the variable **OPCALC** to **REAL**. Then, perform ***identically*** the same operation as before. Figure 10.3 shows the results.

10.5 Conclusion

It can clearly be seen that the Ideal form of the PID algorithm (Non-Interactive PID) creates significantly more noise than the Real form. Ideal PID control leads to unacceptable high wear and tear of physical equipment. This makes Ideal PID unsuitable for field interaction (Non-Interactive). Essentially this means that ideal PID control is unsuitable to operate with real world processes or field values (Valves etc).

10.6 Operation

Again, call up the training application "**General Single Loop with Non-Interactive PID (Ideal Form)**". Set the tuning constants to the same values as before (K=1.1, TINT=0.8 TDER=0.5 and TD=0.05). Then, change SPE from 1000 to 1020 and observe the OP. No changes of the Tuning constants should be made during the whole exercise.

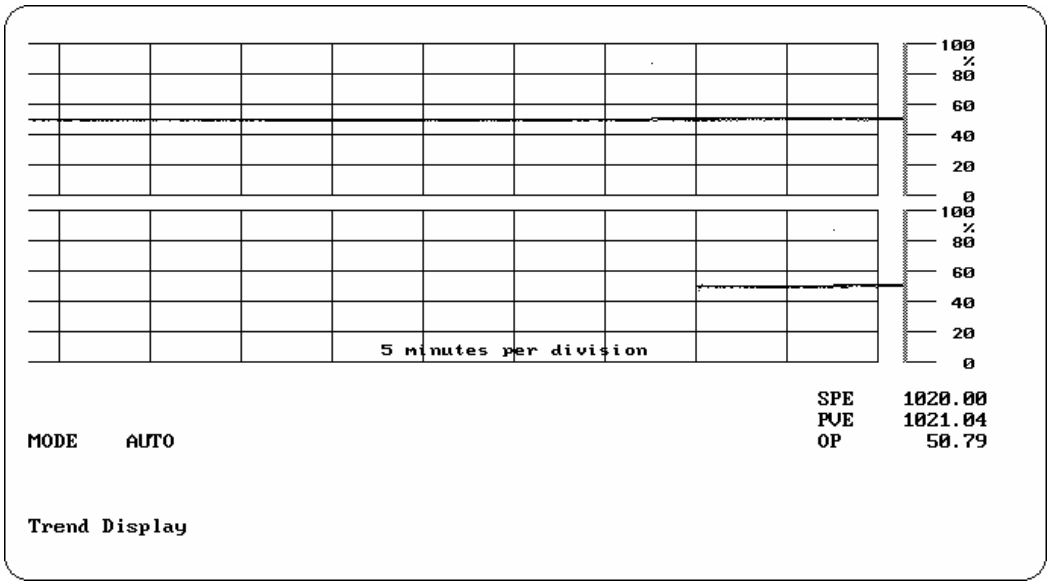


Figure 10.4
Ideal PID Control Action

10.7 Observation

A relative small step function of SPE from 1000 to 1020 results in a derivative action, reaching close to the OPHI limit. The derivative action, as a result of the step function of SPE, takes place for one scan time only (Needle Pulse). It is visible in Figure 10.4 as a single dot of OP close to the value of the OPHI limit, 95% of range. The remaining OP action is based on PID-Control as a result of PVE. It should be noted that this occurs with step functions of SPE or PVE only. In every other case, PID control is performed as a continuous function.

10.8 Conclusion

Step functions of SPE or PVE result in a needle pulse of OP. These pulses are not well received well by field equipment, driven by the output. For example, no valve should experience a single hit by a strong but short pulse of the OP. There is no time for the valve to move and therefore there is no effect on the process. The only result is that the control valve will wear out quickly, when put under such unnecessary high mechanical stress. One important conclusion that comes from this is that you should: Consider the use of Equation Type B or C, where the operator is not able to influence PD Control via step changes of the SPE. The PVE hardly makes step changes.

Discuss the differences and think about possible applications for Ideal and Real PID algorithms. **Remember the use of EQ-Type B and C.** Special attention has to be given to noise. This exercise makes it very clear why the Real PID algorithm is preferred over the Ideal PID algorithm, if used as the Ultimate Secondary Controller (or Field Controller).

Exercise 11

Cascade Control

11.1 Objective

The objective of this exercise is not to provide a guided tour through cascade control, but to introduce the student into the effects of PV tracking, Initialization and Mode changes in Cascade Control. The other aspects of cascade control can easily be explored with the capabilities of this software package. But it is up to the student to experiment.

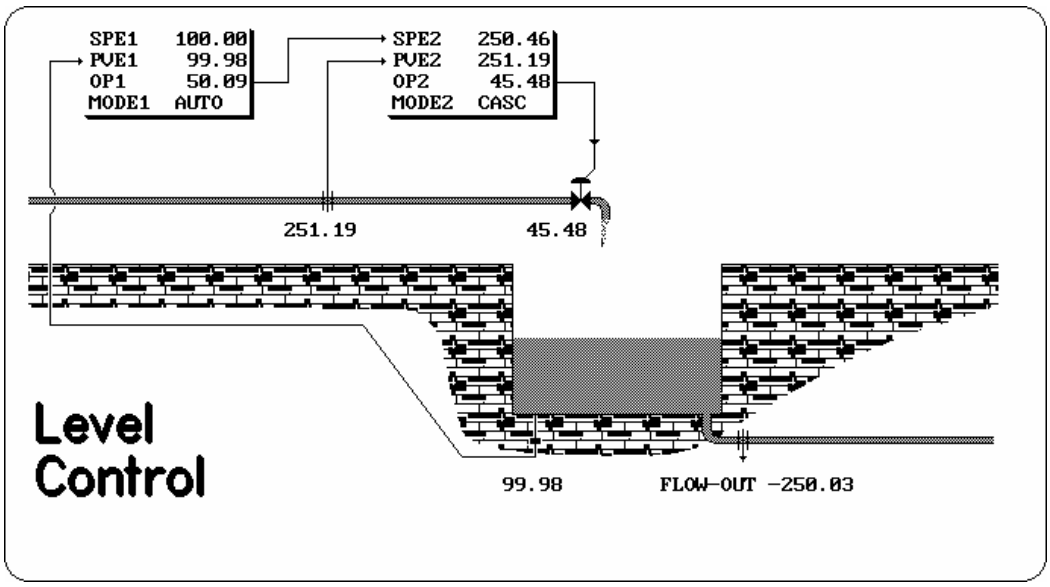


Figure 11.1
Level Control Block Diagram (F2)

11.2 Operation

Call up the training application "**Tank Level Control**" (See Figure Ex.11.1). At this stage no PV tracking or Initialization is active. After the process has settled down, change MODE2 to MANUAL and OP2 to 30% of range. The objective is to continue cascade control when PVE1 has reached approximately a value of 70 (35% of range). As we can see in Figure 11.2 (See left side of the screen), the change of MODE2 and OP2 causes most variables within the cascade control system to drift uncontrolled (OP1, SPE1, SPE2). When PVE1 has reached approximately 70, change SPE1 to 70 and then MODE2 back to CASC.

11.3 Observation

When we change from CASC control to MANUAL (MODE2 from CASC to MANUAL) some variables are unpredictable as discussed earlier. When changing back to CASC control, these unpredictable variables cause a bump within our control system, when they have to change back to real and defined values.

11.4 Conclusion

Based on the observation that unpredictable values of OP1, SPE1 and SPE2 cause a bump in control when we change to CASC control, the following requirements have to be met to avoid this.

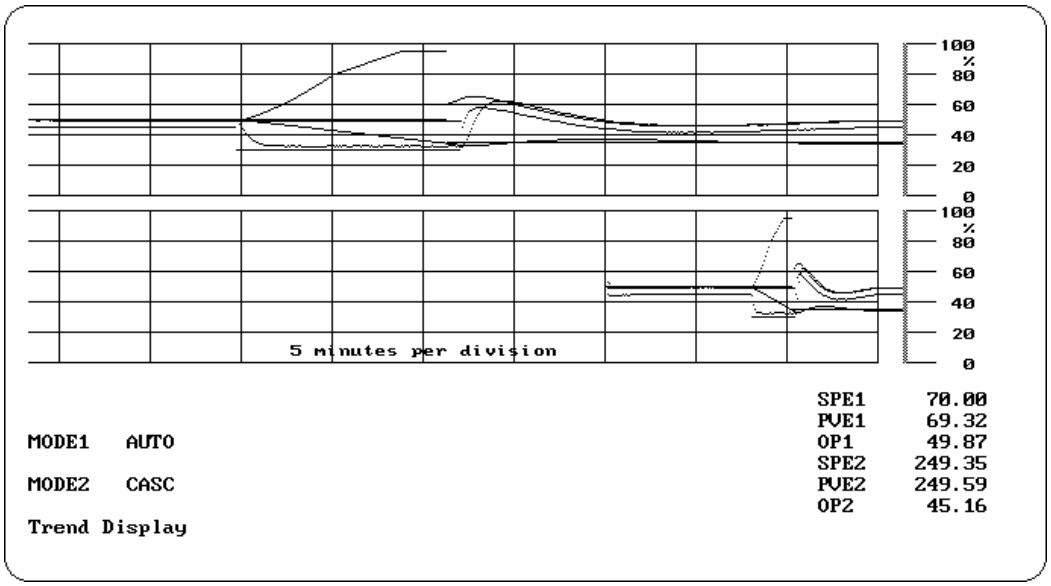


Figure 11.2
Trend Without PV Tracking and Initialization

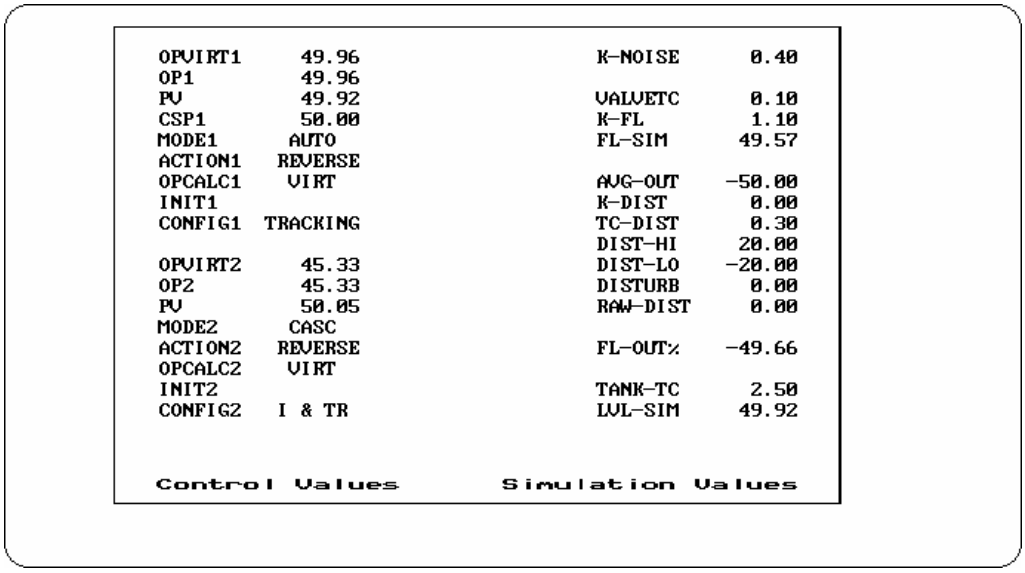


Figure 11.3
Auxiliary Display to Configure CONFIG1 and CONFIG2 (F8)

The secondary controller's (Flow) SPE2 has to follow PVE2 as long as the controller is in MANUAL mode (PV Tracking). The same requirement for PV Tracking has to be made for the primary controller (Level). SPE1 has to follow PVE1. In addition, we cannot permit OP1 to assume any unwanted value when the secondary controller is either in MANUAL or AUTO mode. For this reason OP1 has to assume the same value in % as SPE2 has, as long as the secondary controller is not in CASC mode. This is called Initialization. In our example, the flow controller (Secondary controller) is the initializing controller and has to be configured as such (See Figure 11.3). The level controller is the initialized controller. If the level controller is initialized by the flow controller, the level controller's MODE indication shows for example **I-AUTO**, which means "Initialized with AUTO pending". It is necessary to have the flow controller configured for Initialization and PV Tracking in order to obtain smooth transition from MANUAL (or AUTO) to CASC.

11.5 Operation

In order to satisfy the above requirements, go to display F8 and change **CONFIG1 to TRACKING** and **CONFIG2 to I & TR** (Initialization & Tracking). Change SPE1 to 100 (50% of range), MODE2 to CASC, MODE1 to AUTO and wait until the process has settled down. Change MODE2 to MANUAL. Then change OP2 to 30% of range. When SPE1 has reached approximately 70, change MODE2 back to CASC.

11.6 Conclusion

Based on the observation that unpredictable values of OP1, SPE1 and SPE2 cause a bump in control when we change to CASC control, the previous requirements have to be met to avoid this problem.

11.7 Observation

As seen in Figure 11.4, the values of OP1, SPE1 and SPE2 assume meaningful values at all times when MODE2 is MANUAL. The whole cascade control system is ready to control in CASC mode. No bump occurs at time of transition from MANUAL to CASC (MODE2).

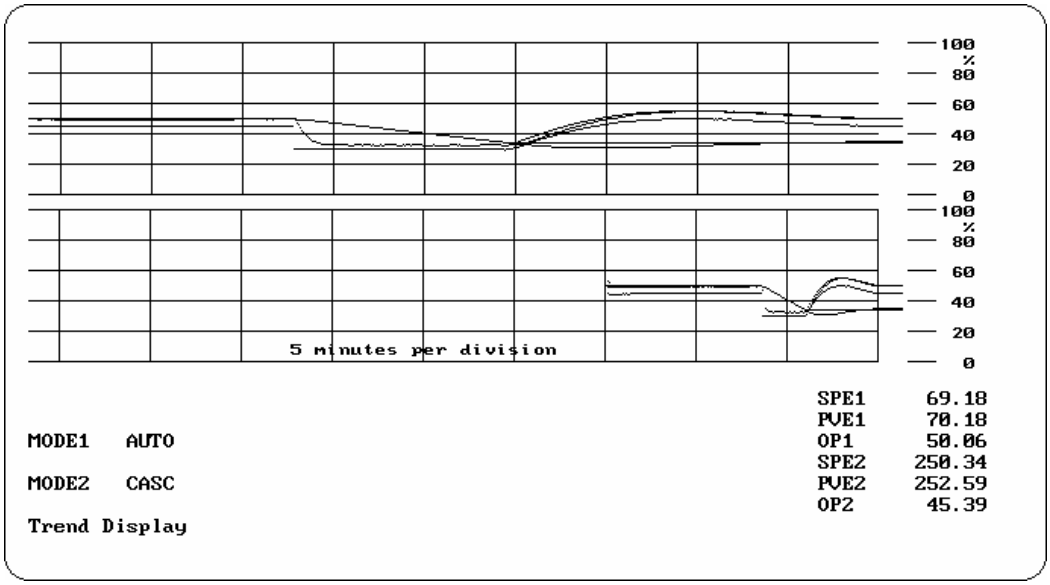


Figure 11.4
Trend Display with PV Tracking and Initialization

11.8 Conclusion

We come to the conclusion, that it is advisable to make use of PV Tracking and Initialization whenever possible. Special care is necessary when deciding on the use of PV Tracking in a primary controller. The incorrect use of PV-Tracking may cause some operational problems. This is explored in more detail in Exercise 13.

Exercise 12

Cascade Control with one Primary and two Secondaries

12.1 Objective

The objective of this exercise is to provide some experience in **Cascade Control with Multiple Secondary Controllers**. Although various aspects are examined in some detail in this section, the software allows the student to explore other areas in more exhaustive detail.

12.2 Training Application

The application used in this exercise deals with cascade control, using two Secondary Controllers and one Primary Controller. In order to drive the setpoints of both Secondary Controllers independently from each other by the same Primary Controller, two independent OP-calculations take place in the Primary Controller. These independent OP-calculations result in two output values OPA and OPB of the Primary Controller and independent modes MODEA and MODEB. See Figure 12.1.

Generally, the amount of data for multiple outputs is too much for one display. As the primary controller in our training application has only two outputs, both are always displayed together on the same display. Viewing both outputs, their modes, alarms and initialization conditions on one screen is more useful in this (teaching) environment.

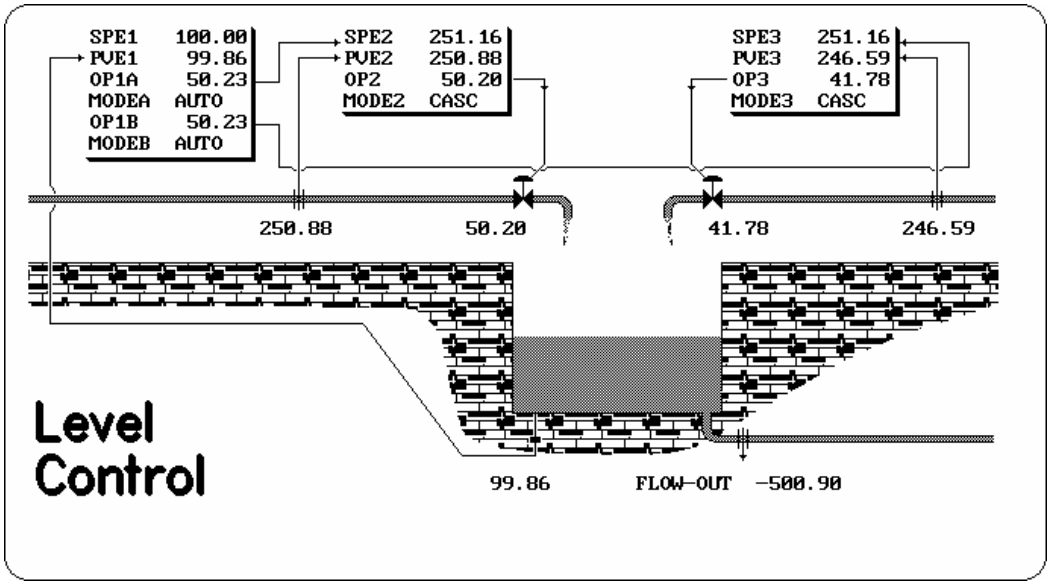


Figure 12.1
Block Diagram

12.3 Operation

Call up the training application "**Tank Level Control with Two Inlets**" .Call the block diagram F2 and study the control concept without changing any values. Then, select trend display F4.

12.4 Observation

The training application is in a fully operational and stable automatic control status and correctly tuned when first retrieved. PVE2 and PVE3, the process variables of the two flow controllers, are approximately equal at the beginning although OP2 and OP3 are not equal. OP2 is approximately 50% and OP3 is approximately 40%. Various reasons for this could include (minor) differences in the supply pressures, valve sizes and pipe diameters.

12.5 Operation

Change MODE3 to MANUAL and then OP3 to 30% of range.

12.6 Observation

We can see that MODEB has changed to I-AUTO and MODEA is still in AUTO. This is an indication that OPB is initialized because MODE3 is in MANUAL. We can actually observe initialization of OPB taking place. OPA is continuing to drive SPE2 and automatic level control still takes place by controlling the flow PVE2 only. As PVE3 has been reduced by manually changing OP3 to 30%, the level starts to drop. The level controller compensates for this by raising OP2 via the flow controller. See Figure 12.2.

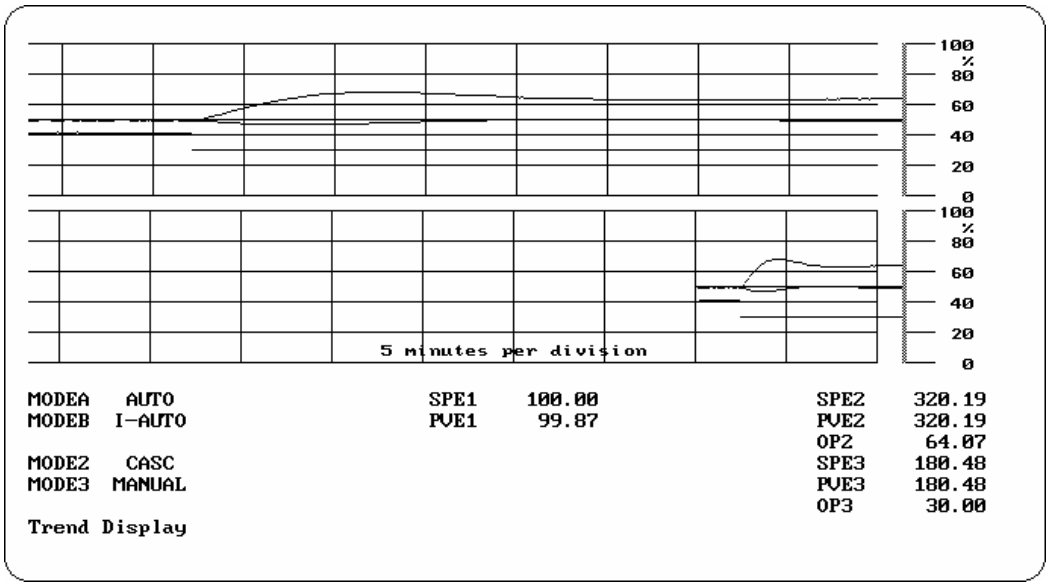


Figure 12.2
Manual OP3 from 40% to 30%

Figures 12.2, 12.3, 12.4 and 12.5 show four trend pens only. These are trends for SPE1, PVE1, OP2 and OP3. The trend pens for SPE2, PVE2, SPE3 and PVE3 have been suspended, using the commands TREND2, TREND3, TREND5 and TREND6. If you require all the trend pens on screen simultaneously when you execute this exercise, then don't use the TRENDx command.

12.7 Operation

Change OP3 even further, from 30% to 10% of range.

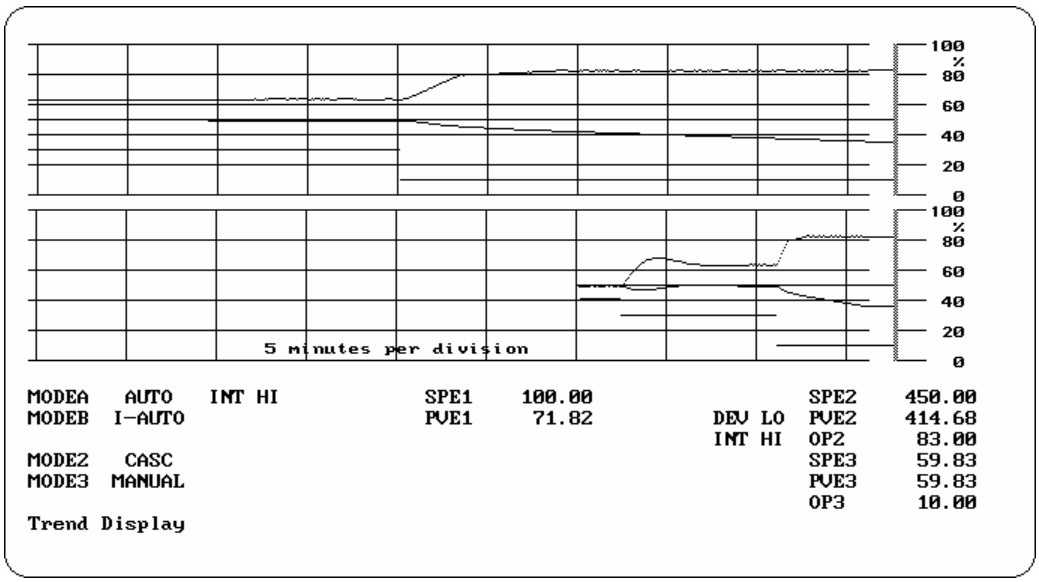


Figure 12.3
Manual OP3 from 30% to 10%

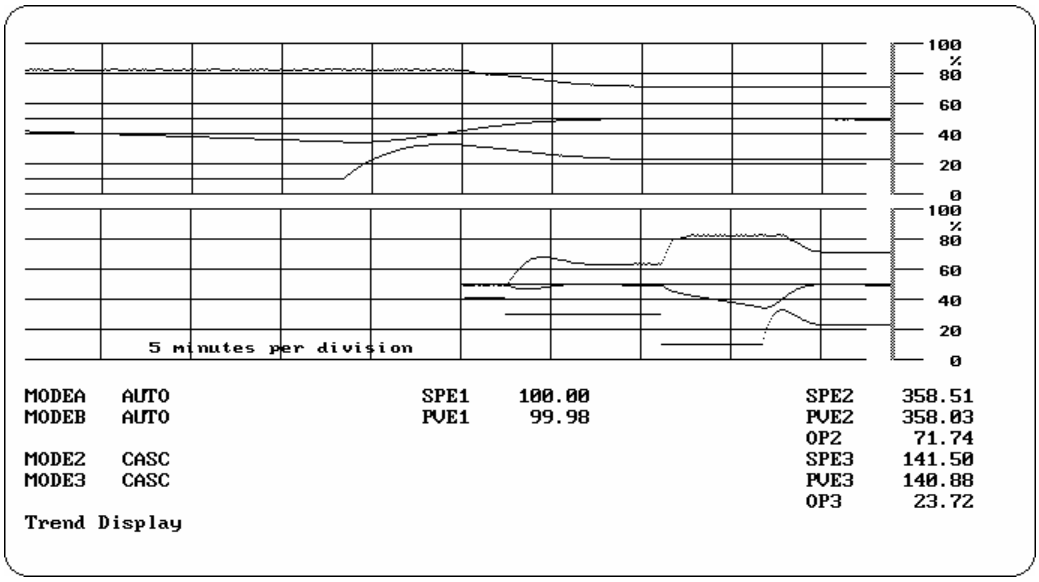


Figure 12.4
Return of MODE3 to CASC

12.8 Observation

The level controller makes an attempt to compensate for the reduction of flow (PVE3) by increasing OP2 via OPA. This attempt is unsuccessful, as OP2 and OPA violate their integral high limits (IHI = 80%). The control action based on one flow controller only, is not strong enough and the tank level drops slowly. In addition, we see an alarm DEV-LO, which tells us that PVE2 deviates too much from SPE2. This alarm has been raised as no integral action can take place beyond the integral limits. Therefore, no control without offset is possible. See Figure 12.3.

12.9 Operation

Change MODE3 to CASC.

12.10 Observation

We can now see that both flow controllers are working together again. As OPA and OPB are calculated independently, OPA is observing the integral limit IHI as long as OPB increases OP3 to a level as to cause OPA to return to within its integral limits. From then on, we can observe that both OP2 and OP3 behave dynamically in an identical manner, as long as no output violates any output limit (including the integral limits). See Figure 12.4

12.11 Operation

Select auxiliary display F8 and set K-DIST to 1.

12.12 Observation

With K-DIST = 1, we have introduced some disturbance. Figure 12.5 shows an example of a trend display where a disturbance has been introduced.

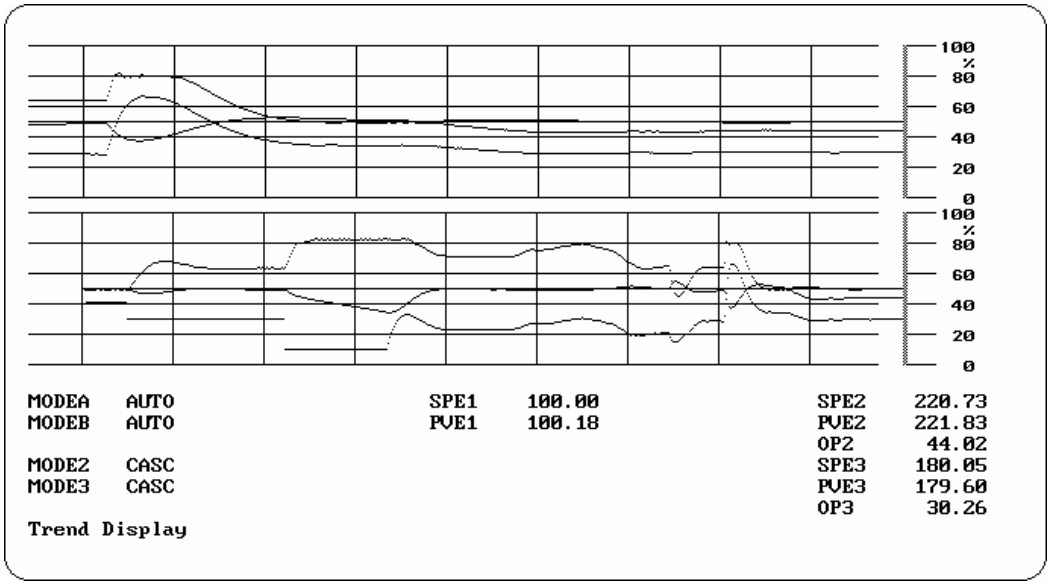


Figure 12.5
Control with Disturbance

12.13 Conclusion

The most important fact that we can glean from this exercise is that automatic control takes place as long as at least one secondary controller was in cascade mode (CASC). It can clearly be seen that both outputs OPA and OPB are calculated totally independently of each other.

Exercise 13

Combined Feedback and Feed Forward Control

13.1 Objective

This exercise is provided to explore combined Feedback and Feed Forward control. The intention of this section is to introduce this control strategy. Although it is not possible to go into all aspects of this topic in this exercise, the control of the training applications have not been simplified. The control examples provided are "Feedheater Control" (Figure 13.1) and "Boiler Level Control" (Figure 13.2).

A few points to remember

The main control is Feed Forward control, whenever all major disturbances are used to calculate Feed Forward control. In most cases, Feedback control serves as a long term correction. In essence this means that it should act against the slow drift of the PV from the setpoint which feed forward cannot be expected to correct. Therefore, Tuning should be done in the following order. Firstly, the flow controller which is common for Feedback and Feed Forward control has to be tuned. Secondly, Feed Forward control and finally Feedback control has to be tuned. The Tuning of Feedback control should aim towards minimum Feedback control action, just enough to eliminate the process drift. Anything more adds to the wear and tear of equipment without significantly improving control results.

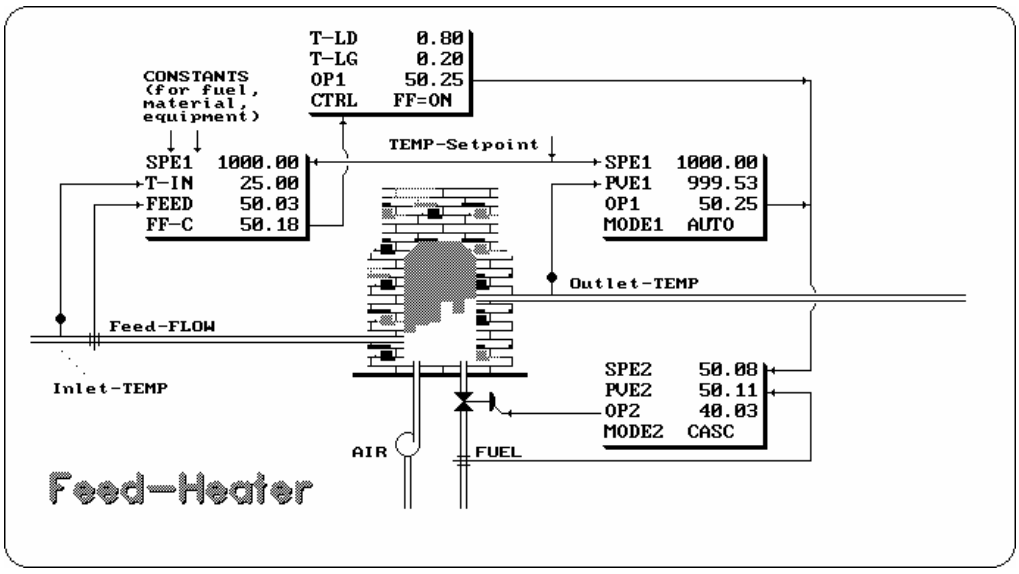


Figure 13.1
Combined Feedback and Feed Forward Control of a Feedheater

13.2 Common Pitfalls

Combined Feed Forward and Feedback Control often makes use of one value in different places. If the operator or engineer is not aware of it, changes of those values may be made with one objective in mind and with a hidden surprise caused by an unknown link.

Feedheater Control Example: SPE1 is used for Feedback control within the Temperature Controller as Setpoint and in addition the same SPE1 is used in Feed Forward control to calculate the temperature difference (SPE1) - (T-IN). The calculation is target temperature minus inlet temperature and is proportional to the amount of fuel required to heat from T-IN to SPE1. This calculation of temperature difference will be corrupted totally, if SPE1 changes unexpectedly because PV-Tracking has been configured for the Temperature Controller.

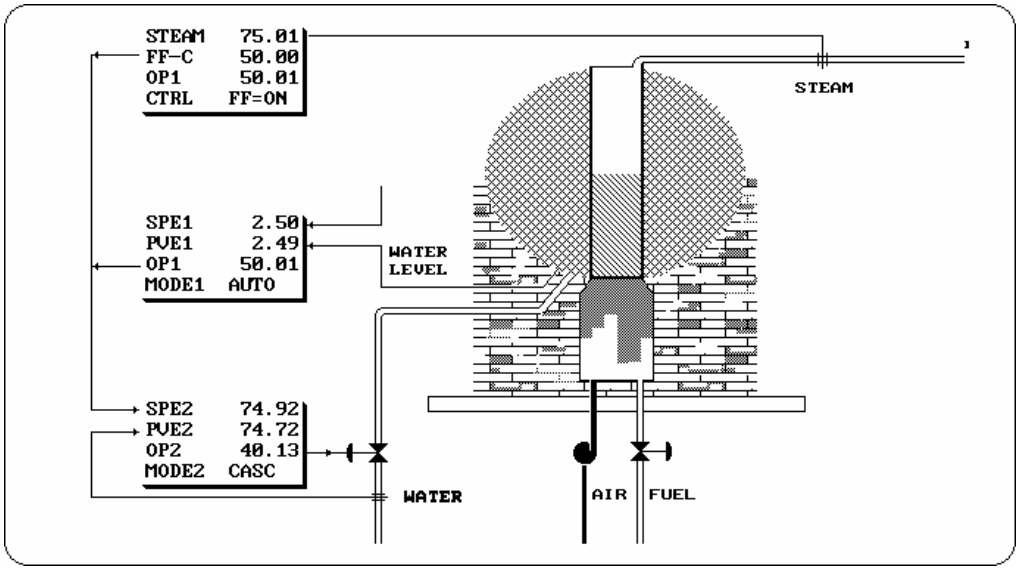


Figure 13.2
Combined Feedback and Feed Forward Control of a Boiler

13.3 Special Effects with Boiler Level Control

The level controller must be tuned for slow control reactions only. Sudden load changes of steam make the water level in boilers appear to change in an opposite direction to that expected. The sudden increase in steam flow decreases boiler pressure. As a result of decreased pressure, the water is boiling more vigorously, which causes the water level to rise. This happens only for short periods of time as a transitional effect, until normal pressure has been restored. If the level controller would react upon those transitional level changes, the control action would go in the wrong direction. A further explanation of this effect is that a sudden increase of steam load causes a sudden increase of evaporation and the bubbles of steam in the water increase. This causes an increase in the water level and a decrease of the water volume in the boiler. Due to this effect, fast flow control should be based on Feed-Forward-Control (Massflow Steam = Massflow Water) and long term level control should be based on Feedback Control only. This is done in effect, if you follow the above advice.

Exercise 14

Dead-Time Compensation in Feedback Control

14.1 Objectives

This exercise will give some practical experience in the Closed Loop Control strategy necessary if a long Dead-Time is part of the process to be controlled. Figure 14.1 shows the block diagram of a control loop with added process simulation as a means of overcoming process Dead-Time.

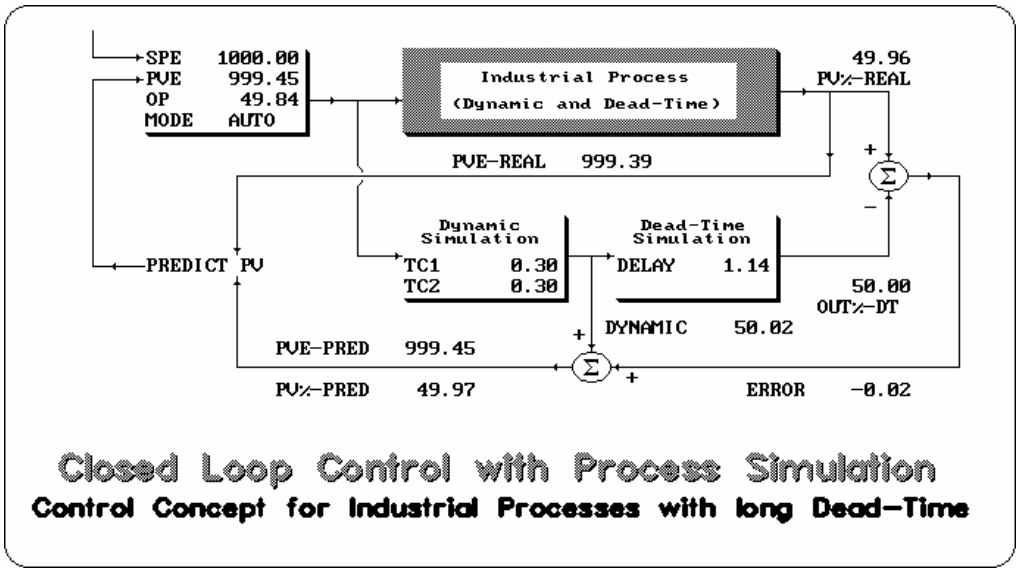


Figure 14.1
Block Diagram

Note Concerning Terminology

Within this exercise, we have a conflict of terminology. As we have no physical industrial plant to control, we have to simulate the behavior of such an industrial plant. If one of the tools of controlling an industrial plant requires a process simulation, we would not know what simulation is meant if we do not make a clear distinction between the process simulation acting as a stand-in on behalf of the real process, and the simulation acting as a tool of control.

Therefore, in the description of this exercise, the term simulation is never used if referring to the real process; the terms for the real industrial process will be used here instead. In this exercise, the term simulation is used only for a process simulation which exists in addition to the industrial process and would exist in a real industrial plant for control purposes as well.

14.2 Operation

Select the training application "**Dead-Time Compensation in Single Loop Control**". The tuning constants for the process simulation are in Detail Display F5. They are K-SIM, BIAS-SIM, TC1, TC2 and DELAY. The values have been initialized to the correct values required to match the real process as close as possible.

Then, select the Trend Display F4 and change MODE to MANUAL. Set the OP to 50% and wait for the process to settle down. Make a change of OP from 50% to 70% of range. See Figure 14.2.

14.3 Observation

Observe the different reaction curves of the process variable coming from the real process (PVE-REAL) and simulated process variable (PVE-PRED). It can be seen that the reaction of the predicted process variable is similar, but if compared to the real process variable there is no dead-time.

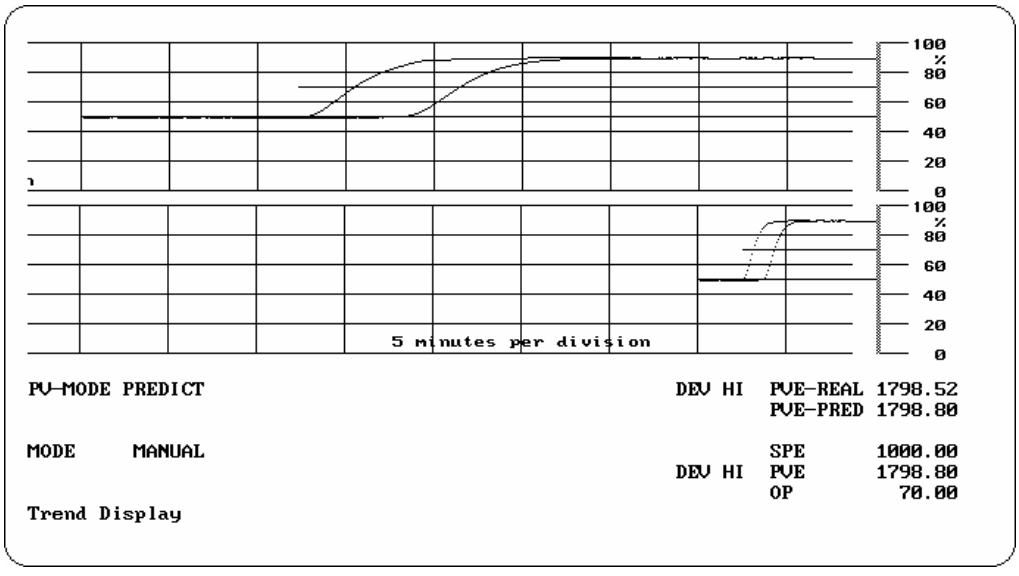


Figure 14.2
PVE-PRED and PVE-REAL Reaction Curves

14.4 Operation

In order to explore the purpose and impact of the simulation tuning constants, do the following for each tuning constant (for K-SIM, BIAS-SIM, TC1, TC2 and DELAY):

- Make sure, the controller is in MANUAL mode.
- Change the tuning constant.
- Make a step change to the OP.
- Carefully observe the reaction curves of PVE-REAL and PVE-PRED.
- Return the tuning constant to its correct value.
- Repeat the above operations with different values for each tuning constant. Avoid experimenting with more than one tuning constant at any one time in order not to confuse the results.

14.5 Observation

The reaction curve of PVE-PRED changes as follows when the tuning constants are changed:

- The change of K-SIM changes the magnitude.
- The change of BIAS-SIM raises or lowers the base line.
- The change of TC1 and TC2 change the dynamic of the simulation (form of reaction curve).
- The change of DELAY causes significant deformations of the reaction curve of PVE-PRED. The characteristic of these deformations indicate whether the dead-time DELAY is too long or too short. Compare the reaction curve obtained with the tuning display F7. See Figure 14.3.

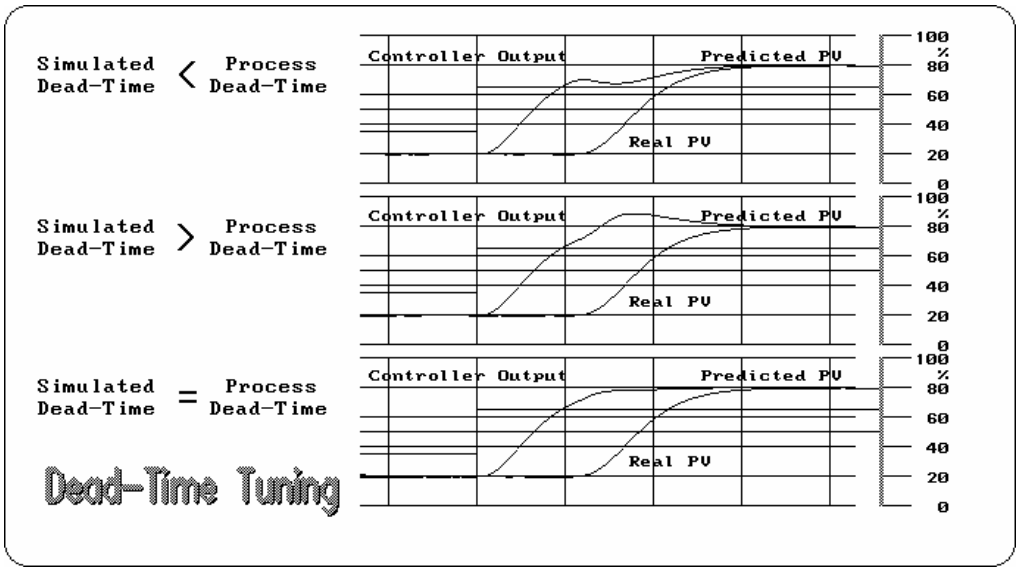


Figure 14.3
Dead-Time Tuning

14.6 Conclusion

K-SIM, BIAS-SIM, TC1 and TC2 shape the simulated reaction curve of PVE-PRED. DELAY serves to match the simulated dead-time with the process dead-time.

14.7 Operation of Simulation Tuning

Select detail display F5 and make sure that the controller is in MANUAL mode. K-SIM, BIAS-SIM, TC1, TC2 and DELAY are the tuning constants for the process simulation. They have to be set to match the real process as close as possible.

Note
The real process can never be matched by simulation with mathematical precision. Therefore, we have to take great care in order to match the simulation as close as possible to the real process. In order to have a realistic training situation, it is not possible to obtain an ideal match within this exercise, but you can come close to it. As a preparation for this tuning exercise, change K-SIM, BIAS-SIM, TC1, TC2 and DELAY to values between 0 and 0.1. This makes sure that the simulation is totally out of tune and tuning is required.

Tuning should be done in the following sequence:

- Use the control loop without simulation and try to tune it using the "Closed Loop Method" according to either Ziegler & Nichols or Pessen. In order to have a classical closed loop without any simulation, change the status variable PV-MODE to REAL. Then, the simulation is still calculated and displayed but is not used for control purposes. This step provides us with a general idea of safe tuning constants (K, TINT and TDER) of the closed loop.
- Switch the controller mode to MANUAL. Make step changes to the OP and experiment with the tuning constants K-SIM and BIAS-SIM until the correct values have been found. The values for K-SIM and BIAS-SIM are correct if the final value of PVE-REAL and PVE-PRED approach the same value without bias towards the end of both reaction curves
- Manipulate TC1 and TC2 in order to match as closely as possible the dynamic behavior of both reaction curves. The observation of the reaction curve of PVE-PRED is very difficult because good judgment is necessary to separate the effect of the dynamic changes from the effect of non-matching dead-times.
- Match the dead-time of the real industrial process and the simulation. Make use of the tuning display F7. Repeat the preceding step and this step as often as necessary to obtain an optimum match of dynamic and dead-time.
- Change the status variable PV-MODE to PREDICT. This makes the controller use the predicted variable PVE-PRED instead of PVE-REAL. Obtain optimum tuning constants for closed loop control using the predicted variable PVE-PRED. You may either use the method according to Ziegler & Nichols or Pessen. Detail display F3 contains the necessary tuning constants.
- Even though the stability problem of the closed loop is dramatically reduced, it still migrates to some extent into the predicted PVE-PRED via the error calculation of the control strategy. The magnitude of this impact can be subjected to a complex mathematical evaluation but is in practice quite unpredictable. The reason therefore, lies mainly in the unaccountable differences between process and simulation.

  Hence, in order to make sure that the problems discussed above don't cause unpleasant and unexpected surprises, a long time of observation of any unexpected stability problems is essential. As a guide, the time of observation should be at least discussed above twenty times the dead-time.
- Based on the observation made in the preceding step, the tuning constant K has to be lowered to a point where smooth control is assured.

14.8 Conclusion

The basic idea of Dead-time compensation is straight forward. Nevertheless, it is important to realize that it is a strategy of intermeshed loops. The control loop using the predicted PVE-PRED is intermeshed with remnants of the real PVE-REAL via the error calculation. In addition, one has to realize that the reaction of control actions is predicted only. No prediction whatsoever can be made about the impact of disturbances. Derivative control is most effective in counteracting integral and lagging behavior, but has its limitations if applied to dead-time problems. Therefore, one should refrain from using PID-Control and accept that PI-Control is the preferred solution in these cases.

Exercise 15

Static Value Alarm

15.1 Objective

The objective of this exercise is to demonstrate the use of statistics is process control. In some cases of pressure measurement, a pressure meter ceases to provide correctly updated values. The last correct value obtained, is provided continuously as a static value which may still be within its correct range and not be recognized by either operator or automatic alarm mechanisms as being an incorrect value. This exercise makes use of a continuous (running) standard deviation calculation. An alarm will be raised whenever no changes of the measured value take place. The concept of detection of a false value is based on the premise that the standard deviation value is too low. See Figure 15.1.

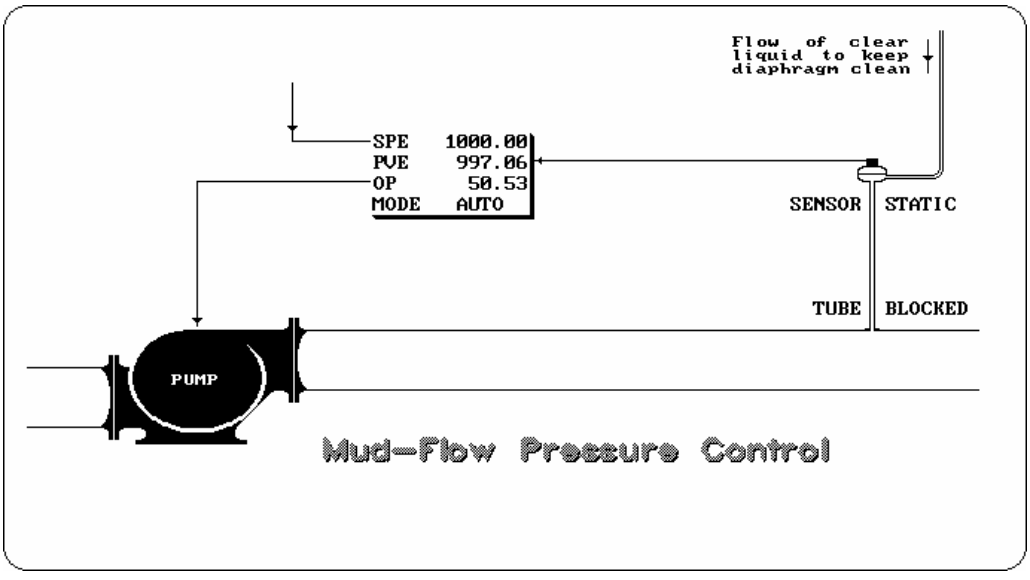


Figure 15.1
Block Diagram

15.2 Operation

Select the training application "**Pressure Control with Static-Sensor-Alarm**". In order to observe the operation of the static sensor alarm, select display F2 and simulate a blockage of the sensor pipe. To simulate a blocked or free sensor pipe, manipulate the status variable TUBE. TUBE can be set to either status FREE or BLOCKED. See Figure 15.1.

15.3 Observation

When the sensor pipe is blocked, the status variable SENSOR displays the alarm STATIC. Immediately after removing the blockage, the alarm disappears and accurate readings of PVE are available.

15.4 Conclusion

This example demonstrates that it is worthwhile to explore the use of statistical means in order to improve some aspects of process control.